Corollary 9.23: *If $\sum_1^\infty f_k$ converges uniformly to f on an interval I and if each of the functions f_k is continuous on I, then f is continuous on I.*

Proof. This follows immediately from Theorem 9.22. ■

9.4.1 Dini's Theorem

✂ *Advanced section. May be omitted.*

Observe that Theorem 9.22 provides a *sufficient* condition for continuity of the limit function f. The condition is not necessary. (The sequence in Example 9.6 converges to the zero function, which is continuous, even though the convergence is not uniform.)

Under certain circumstances, however, uniform convergence *is* necessary, as Theorem 9.24 shows. (See also Exercise 9.4.6.) This theorem is due to Ulisse Dini (1845–1918) and gives a condition under which pointwise convergence of a sequence of continuous functions to a continuous function must be uniform.

Theorem 9.24 (Dini) *Let $\{f_n\}$ be a sequence of continuous functions on an interval $[a,b]$. Suppose for each $x \in [a,b]$ and for all $n \in \mathbb{N}$,*

$$f_n(x) \geq f_{n+1}(x).$$

Suppose in addition that for all $x \in [a,b]$

$$f(x) = \lim_n f_n(x).$$

If f is continuous, then the convergence is uniform.

Proof. Suppose the convergence were *not* uniform. Then

$$\max_{x\in[a,b]} (f_n(x) - f(x))$$

does not approach zero as $n \to \infty$ (see Exercise 9.3.12). Hence there exists $c > 0$ such that for infinitely many $n \in \mathbb{N}$,

$$\max_{x\in[a,b]} (f_n(x) - f(x)) > c > 0.$$

Now, for each $n \in \mathbb{N}$, $f_n - f$ is continuous, so it achieves a maximum value at a point $x_n \in [a,b]$. By the Bolzano-Weierstrass theorem we can thus choose a subsequence $\{x_{n_k}\}$ of the sequence $\{x_n\}$ such that $\{x_{n_k}\}$ converges to a point $x_0 \in [a,b]$. Note that we must have

$$f_{n_k}(x_{n_k}) - f(x_{n_k}) > c$$

for all $k \in \mathbb{N}$.

Because of our assumption that $f_n(x) \geq f_{n+1}(x)$ for all $n \in \mathbb{N}$ and $x \in [a,b]$, we infer

$$f_n(x_{n_k}) - f(x_{n_k}) > c \text{ for each } n \leq n_k.$$

Now fix n and let $k \to \infty$. Using the continuity of the functions $f_n - f$ we obtain $f_n(x_0) - f(x_0) \geq c$ for all $n \in \mathbb{N}$. But this is impossible since $f_n(x_0) \to f(x_0)$ by hypothesis. Thus our assumption that the convergence is not uniform has led to a contradiction. ■

Example 9.25: The sequence of continuous functions $f_n(x) = x^n$ is converging monotonically to a function f on the interval $[0,1]$. But that function f is (as we have seen before) discontinuous at $x = 1$, so immediately we know that the convergence cannot be uniform. Dini's theorem implies that the convergence is uniform on $[0,b]$ for any $0 < b < 1$ since the function f is continuous there. ◀

Exercises

9.4.1 Can a sequence of discontinuous functions converge uniformly on an interval to a continuous function?

9.4.2 Let $f_n(x) = x^n$, $0 \leq x \leq 1$. Try to imitate the proof of Theorem 9.22 for $x_0 = 1$ and observe where the proof breaks down.

9.4.3 Let $\{f_n\}$ be a sequence of functions each of which is uniformly continuous on an open interval (a,b). If $f_n \to f$ uniformly on (a,b) can you conclude that f is also uniformly continuous on (a,b)?

9.4.4 Give an example of a sequence of continuous functions $\{f_n\}$ on the interval $(0,1)$ that is monotonic decreasing and converges pointwise to a continuous function f on $(0,1)$ but for which the convergence is not uniform. Why does this not contradict Theorem 9.24?

9.4.5 Give an example of a sequence of continuous functions $\{f_n\}$ on the interval $[0,\infty)$ that is monotonic decreasing and converges pointwise to a continuous function f on $[0,\infty)$ but for which the convergence is not uniform. Why does this not contradict Theorem 9.24?

9.4.6 Let $\{f_n\}$ be a sequence of continuous nondecreasing functions defined on an interval $[a,b]$. Suppose $f_n \to f$ pointwise on $[a,b]$. Prove that if f is continuous on $[a,b]$, then the convergence is uniform. Observe that, in this exercise, the *functions* are assumed monotonic, whereas in Theorem 9.24 it is the *sequence* that we assume monotonic.

9.4.7 The proof of Theorem 9.24 depends on the compactness of the interval $[a,b]$. The compactness argument used here relied on the Bolzano-Weierstrass theorem. Attempt another proof using one of our other strategies from Section 4.5.

9.4.8 Prove this variant on Dini's theorem (Theorem 9.24). Let $\{f_n\}$ be a sequence of continuous functions on an interval $[a,b]$. Suppose for each $x \in [a,b]$ and for all $n \in \mathbb{N}$, $f_n(x) \leq f_{n+1}(x)$. Suppose in addition that for all $x \in [a,b]$ $\lim_n f_n(x) = \infty$. Show that for all $M > 0$ there is an integer N so that $f_n(x) > M$ for all $x \in [a,b]$ and all $n \geq N$. Show that this conclusion would not be valid without the monotonicity assumption.

9.4.9 Show that if, in Exercise 9.4.8, the interval $[a,b]$ is replaced by the unbounded interval $[0,\infty)$ or the nonclosed interval $(0,1)$, then the conclusion need not be valid.

9.4.10 Let $\{f_n\}$ be a sequence of Lipschitz functions on $[a,b]$ with common Lipschitz constant M. (This means that $|f_n(x) - f_n(y)| \leq M|x-y|$ for all $n \in \mathbb{N}$, $x, y \in [a,b]$.)

(a) If $f = \lim_n f_n$ pointwise, then f is continuous and, in fact, satisfies a Lipschitz condition with constant M.

(b) If $f = \lim_n f_n$ pointwise the convergence is uniform.

(c) Show by example that the results in (a) and (b) fail if we weaken our hypotheses by requiring only that each function is a Lipschitz function. (Here, the constant M may depend on n.)

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9.4.11 (Continuous convergence and uniform convergence) A sequence of functions $\{f_n\}$ defined on an interval I is said to *converge continuously* to the function f if $f_n(x_n) \to f(x_0)$ whenever $\{x_n\}$ is a sequence of points in the interval I that converges to a point x_0 in I. Prove the following theorem:

> *Let $\{f_n\}$ be a sequence of continuous functions on an interval $[a,b]$. Then $\{f_n\}$ converges continuously on $[a,b]$ if and only if $\{f_n\}$ converges to f uniformly on $[a,b]$.*

Does the theorem remain true if the interval $[a,b]$ is replaced with (a,b) or $[a,\infty)$?

9.4.12 Show that the sequence $f_n(x) = x^n/n$ converges uniformly on [0,1]:

(a) By direct computation using the definition of uniform convergence

(b) By using Theorem 9.24

(c) By using Exercise 9.4.6

(d) By using Exercise 9.4.11

9.5 Uniform Convergence and the Integral

Calculus students often learn the following simple computation. The geometric series

$$\frac{1}{1-t} = \sum_{k=0}^{\infty} t^k \tag{10}$$

is valid on the interval $(-1,1)$. An integration of both sides for t in the interval $[0,x]$, and any choice of $x < 1$ will yield

$$-\log(1-x) = \int_0^x \frac{1}{1-t}\,dt = \sum_{k=0}^{\infty} \frac{x^{k+1}}{k+1}.$$

Indeed this identity is valid and provides a series expansion for the logarithm function. But can this really be justified?

In general, do we know that if $f(x) = \sum_0^\infty f_n(x)$ on an interval $[a,b]$, then

$$\int_a^b f(x)\,dx = \sum_0^\infty \int_a^b f_n(x)\,dx?$$

In fact, we already observed in Section 9.3 that during the early part of the nineteenth century, some prominent mathematicians took for granted the permissibility of term by term integration of convergent infinite series of functions. This was true of Fourier, Cauchy, and Gauss. Example 9.6, cast in the setting of sequences of integrable functions, shows that these mathematicians were mistaken.

9.5.1 Sequences of Continuous Functions

Around the middle of the nineteenth century, Weierstrass showed that term by term integration is permissible when the series of integrable functions converges *uniformly*. Let us first verify this result for sequences of continuous functions.

Theorem 9.26: *Suppose that $f(x) = \lim_{n\to\infty} f_n(x)$ for all $x \in [a,b]$, that each function f_n is continuous on $[a,b]$, and that the convergence is uniform. Then*

$$\int_a^b f(x)\,dx = \lim_{n\to\infty} \int_a^b f_n(x)\,dx.$$

Proof. By Theorem 9.22, f is continuous, so $\int_a^b f(x)\,dx$ exists. We must show that $\int_a^b f_n(x)\,dx \to \int_a^b f(x)\,dx$.

Let $\varepsilon > 0$. We wish to obtain $N \in \mathbb{N}$ such that

$$\left| \int_a^b f_n(x)\,dx - \int_a^b f(x)\,dx \right| < \varepsilon \text{ for all } n \geq N.$$

We calculate that for any $n \in \mathbb{N}$

$$\left| \int_a^b f_n(x)\,dx - \int_a^b f(x)\,dx \right| = \left| \int_a^b [f_n(x) - f(x)]\,dx \right|$$

$$\leq \int_a^b |f_n(x) - f(x)|\,dx \leq \int_a^b \max_{x\in[a,b]} |f_n(x) - f(x)|\,dx$$

$$\leq (b-a)\left(\max_{x\in[a,b]} |f_n(x) - f(x)| \right).$$

Since $f_n \to f$ uniformly on $[a,b]$, there exists $N \in \mathbb{N}$ such that

$$\max_{x\in[a,b]} |f_n(x) - f(x)| < \frac{\varepsilon}{b-a} \text{ for all } n \geq N.$$

Thus, for $n \geq N$, we have

$$\left| \int_a^b f_n(x)\,dx - \int_a^b f(x)\,dx \right| \leq (b-a)\frac{\varepsilon}{b-a} = \varepsilon$$

as was to be shown. ■

Applying the theorem to the partial sums S_n of a series allows us to express this result for series.

Corollary 9.27: *If an infinite series of continuous functions $\sum_0^\infty f_k$ converges uniformly to a function f on an interval $[a,b]$, then f is also continuous and*

$$\int_a^b f(x)\,dx = \sum_0^\infty \int_a^b f_k(x)\,dx.$$

Example 9.28: Let us justify the computations that we made in our introduction to this topic. The geometric series

$$\frac{1}{1-t} = \sum_{k=0}^\infty t^k \tag{11}$$

converges pointwise on the interval $(-1,1)$. Let $0 < x < 1$. By the M-test we see that this series converges uniformly on $[0,x]$. Each of the terms in the sum is continuous. As a result we may apply our theorem to integrate term by term just as we might have seen in a calculus course. Thus

$$\int_0^x \frac{1}{1-t}\,dt = \sum_{k=0}^\infty \frac{x^{k+1}}{k+1}.$$

◀

9.5.2 Sequences of Riemann Integrable Functions

✂ *Enrichment section. May be omitted.*

In Theorem 9.26 we required that the functions f_n be continuous. Suppose we now weaken our hypotheses for these functions by requiring only that they be integrable, but still requiring the sequence $\{f_n\}$ to converge uniformly to f. We note that in all respects the proof is the same. Thus, if a uniformly convergent sequence of integrable functions converges to an integrable function, we can integrate the sequence term by term. Our next theorem shows that a uniform limit of integrable functions must be integrable and so we have the following extension of Theorem 9.26.

Theorem 9.29: *Let $\{f_n\}$ be a sequence of functions Riemann integrable on an interval $[a,b]$. If $f_n \to f$ uniformly on $[a,b]$, then f is Riemann integrable on $[a,b]$ and*

$$\int_a^b f(x)\,dx = \lim_n \int_a^b f_n(x)\,dx.$$

Proof. The proof of Theorem 9.26 can be repeated if we happen to know that f is integrable. Thus we need only show that the limit function f is integrable on $[a,b]$.

One proof (see Exercise 9.5.7) would be to show that f is bounded and continuous everywhere except at a set of measure zero. It follows by Theorem 8.17 that f is Riemann integrable.

We can also give a proof by constructing, for any $\varepsilon > 0$, step functions having the property of Exercise 8.6.5. Since this proof is one that was available to nineteenth-century mathematicians, who would not have known about sets of measure zero, this is worth presenting, if only for historical reasons.

Let $\varepsilon > 0$. We wish to find step functions L and U such that

$$L(x) \le f(x) \le U(x)$$

for all $x \in [a,b]$, with

$$\int_a^b [U(x) - L(x)]\,dx < \varepsilon.$$

We shall obtain the functions L and U in three natural steps:

1. We approximate f by one of the functions f_N.
2. We obtain corresponding step functions L_N and U_N approximating f_N.
3. We modify L_N and U_N to obtain L and U.

We proceed according to the aforementioned plan.

(i) Since $f_n \to f$ *uniformly*, there exists $N \in \mathbb{N}$ such that

$$|f_N(x) - f(x)| \le \frac{\varepsilon}{4(b-a)} \text{ for all } x \in [a,b].$$

(ii) Since f_N is integrable by hypothesis, there exist step functions L_N and U_N such that

$$L_N(x) \le f_N(x) \le U_N(x)$$

for all $x \in [a,b]$ and

$$\int_a^b [U_N(x) - L_N(x)]\,dx < \frac{\varepsilon}{2}.$$

(iii) Let us define the step functions U and L by

$$L(x) = L_N(x) - \frac{\varepsilon}{4(b-a)}\ ,\ U(x) = U_N(x) + \frac{\varepsilon}{4(b-a)}$$

for all $x \in [a,b]$.

We then have

$$L(x) < L_N(x) - |f(x) - f_N(x)| \le f(x) \le U_N(x) + |f(x) - f_N(x)| < U(x)$$

and

$$\begin{aligned}\int_a^b [U(x) - L(x)]\,dx &= \int_a^b \left\{ [U_N(x) - L_N(x)] + \frac{\varepsilon}{2(b-a)} \right\} dx \\ &= \int_a^b [U_N(x) - L_N(x)]\,dx + \int_a^b \frac{\varepsilon}{2(b-a)}\,dx \\ &< \frac{\varepsilon}{2} + \frac{\varepsilon}{2} = \varepsilon,\end{aligned}$$

as was to be shown. ■

Corollary 9.30: *If an infinite series of integrable functions $\sum_0^\infty f_k$ converges uniformly to a function f on an interval $[a,b]$, then f is also integrable and*

$$\int_a^b f(x)\,dx = \sum_0^\infty \int_a^b f_k(x)\,dx.$$

Example 9.31: Let $f_n(x) = e^{-nx^2}$. Then for each $x \in [1,2]$ and for every $n \in \mathbb{N}$, $0 < e^{-nx^2} \le e^{-n}$ and $e^{-n} \to 0$, so $f_n \to 0$ uniformly on [1,2]. It follows that

$$\lim_n \int_1^2 e^{-nx^2}\,dx = \int_1^2 0\,dx = 0.$$

◄

Note. We end this section with a short note that considers whether our main theorem would be true under weaker hypotheses than uniform convergence.

It is possible for a sequence $\{f_n\}$ of functions to converge pointwise (but *not* uniformly) to a function f on $[a,b]$ and still have

$$\lim_n \int_a^b f_n(x)\,dx = \int_a^b f(x)\,dx. \tag{12}$$

For example, suppose we modify the functions of Example 9.6 so that

$$f_n(1/(2n)) = 1$$

instead of $f_n(1/(2n)) = 2n$. We still have $f_n \to 0$ pointwise (but not uniformly), but now

$$\int_0^1 f_n(x)\,dx \to 0.$$

These functions form a *uniformly bounded* sequence of functions : that is, there exists a constant M ($M = 1$ in this case) such that $|f_n(x)| \le M$ for all $n \in \mathbb{N}$ and all $x \in [0,1]$. A theorem (whose proof is beyond the scope of this chapter) asserts that if a uniformly bounded sequence of integrable functions $\{f_n\}$ converges pointwise to

an integrable function f on $[a,b]$, then the identity (12) holds. We cannot drop the hypothesis of integrability of f in this theorem. If, for example, $\{r_n\}$ is an enumeration of the rationals in $[0,1]$ and

$$f_n(x) = \begin{cases} 1, & \text{if } x = r_1, r_2, \ldots, r_n \\ 0, & \text{otherwise,} \end{cases}$$

then

$$\lim_{n\to\infty} f_n(x) = f(x) = \begin{cases} 1, & \text{if } x \in \mathbb{Q} \cap [0,1] \\ 0, & \text{if } x \in [0,1] \setminus \mathbb{Q} \end{cases}$$

and f is not integrable on $[0,1]$ by Exercise 9.2.3.

9.5.3 Sequences of Improper Integrals

Thus far we have studied limits of ordinary integrals of functions on a finite interval $[a,b]$. What if the integrals are to be taken on an infinite interval?

More narrowly, let us just ask for the validity of the formulas:

$$\lim_{n\to\infty} \int_a^\infty f_n(t)\,dt = \int_a^\infty f(t)\,dt$$

in case $f_n \to f$ or

$$\sum_{k=1}^\infty \int_a^\infty g_k(t)\,dt = \int_a^\infty f(t)\,dt$$

in case $f = \sum_{k=1}^\infty g_k$. A fast and glib answer would be that we hardly expect these to be true for pointwise convergence but certainly uniform convergence will suffice.

But these integrals involve an extra limit operation and we therefore need extra caution. Indeed, the following example shows that uniform convergence is far from enough. It is not just the "smoothness" of the convergence that is an issue here.

Example 9.32: Let $f_n(x)$ be defined as $1/n$ for all values of $x \in [0,n]$ but as zero for $x > n$. Then the sequence $\{f_n\}$ converges to zero uniformly on the interval $[0,\infty)$. But the integrals do not converge to zero (as we would have hoped) since

$$\int_0^\infty f_n(t)\,dt = 1$$

for all n. ◀

What further condition can we impose so that, together with uniform convergence, we will be able to take the limit operation inside the integral

$$\lim_{n\to\infty} \int_0^\infty f_n(t)\,dt?$$

The condition we impose in the theorem just requires that all the functions are controlled or dominated by some function that is itself integrable. In Example 9.32 note that there is no possibility of an integrable function g on $[0,\infty)$ such that $f_n(x) \le g(x)$ for all n and x. Theorems of this kind are called *dominated convergence theorems*.

Theorem 9.33: *Let $\{f_n\}$ be a sequence of continuous functions on the interval $[a,\infty)$ such that $f_n \to f$ uniformly on any interval $[a,b]$. Suppose that there is a continuous function g on $[a,\infty)$ such that*

$$|f_n(x)| \leq g(x)$$

for all $a \leq x$ and all n. Suppose that the integral $\int_a^\infty g(x)\,dx$ exists. Then

$$\lim_{n\to\infty} \int_a^\infty f_n(t)\,dt = \int_a^\infty f(t)\,dt.$$

Proof. As a first step let us show that f is integrable on $[a,\infty)$. Certainly f is continuous since it is a uniform limit of a sequence of continuous functions. Since each $|f_n(x)| \leq g(x)$ it follows that $|f(x)| \leq g(x)$. We check then

$$\left| \int_c^d f(t)\,dt \right| \leq \int_c^d |f(t)|\,dt \leq \int_c^d g(t)\,dt.$$

Since g is integrable, it follows by the Cauchy criterion for improper integrals (see Exercise 8.5.11) that the integral $\int_c^d g(t)\,dt$ can be made arbitrarily small for large c and d. But then so also is the integral $\int_c^d f(t)\,dt$, and a further application of the Cauchy criterion for improper integrals shows that f is integrable. (Indeed this argument shows that f is absolutely integrable in fact.)

Now let $\varepsilon > 0$. Choose L_0 so large that

$$\int_{L_0}^\infty g(t)\,dt < \varepsilon/4.$$

Choose N so large that

$$|f_n(t) - f(t)| < \frac{\varepsilon}{2(L_0 - a)}$$

if $n \geq N$ and $t \in [a, L_0]$. This is possible because $f_n \to f$ uniformly on $[a, L_0]$. Then we have

$$\left| \int_a^\infty f_n(t)\,dt - \int_a^\infty f(t)\,dt \right| \leq \int_a^{L_0} |f_n(t) - f(t)|\,dt + \int_{L_0}^\infty 2g(t)\,dt$$

$$\leq \frac{\varepsilon}{2(L_0 - a)}(L_0 - a) + \frac{2\varepsilon}{4} = \varepsilon$$

for all $n \geq N$. This proves the assertion of the theorem. ■

Exercises

9.5.1 Prove that

$$\lim_{n\to\infty} \int_{\frac{\pi}{2}}^{\pi} \frac{\sin nx}{nx}\,dx = 0.$$

9.5.2 Prove that

$$\int_0^\pi \sum_{n=1}^\infty \frac{\sin nx}{n^2}\,dx = \sum_{n=1}^\infty \frac{2}{(2n-1)^3}.$$

9.5.3 Show that if $f_n \to f$ uniformly on $[a,b]$ and each f_n is continuous then the sequence of functions

$$F_n(x) = \int_a^x f_n(t)\,dt$$

also converges uniformly on $[a,b]$.

9.5.4 Show that if $f_n \to f$ uniformly on $[a,b]$ and each f_n is continuous then

$$\lim_{n\to\infty} \int_a^b \left(\int_a^x f_n(t)\,dt \right) dx = \int_a^b \left(\int_a^x f(t)\,dt \right) dx.$$

9.5.5 Show that the series

$$\sum_{k=0}^{\infty} \frac{x^k}{k!}$$

converges uniformly on $[-a,a]$ for every $a \in \mathbb{R}$ but does not converge uniformly on all of the real line. (Does it converge pointwise on the real line?) Obtain a series representation for

$$\int_{-a}^{a} \sum_{k=0}^{\infty} \frac{x^k}{k!}\,dx.$$

9.5.6 Let $\{f_n\}$ be a sequence of continuous functions on an interval $[a,b]$ that converges uniformly to a function f. What conditions on g would allow you to conclude that

$$\lim_{n\to\infty} \int_a^b f_n(t)g(t)\,dt = \int_a^b f(t)g(t)\,dt?$$

9.5.7 Let $\{f_n\}$ be a sequence of bounded functions each continuous on an interval $[a,b]$ except at a set of measure zero. Show that if $f_n \to f$ uniformly on $[a,b]$, then the function f is also bounded and continuous on $[a,b]$ except at a set of measure zero. Conclude that a uniformly convergent sequence of Riemann integrable functions must converge to a function that is also Riemann integrable.

9.5.8 Let $p > -1$. Show that

$$\lim_{n\to\infty} \int_1^n \left(1 - \frac{t}{n}\right)^n t^p\,dt = \int_1^{\infty} e^{-t}t^p\,dt.$$

9.5.9 Formulate and prove a version of the dominated convergence theorem (Theorem 9.33) that would apply to improper integrals on an interval $[a,b]$ where the point of unboundedness is at the endpoint a.

9.5.10 Compute the limit

$$\lim_{n\to\infty} \int_0^1 \frac{e^{-nt}}{\sqrt{t}}\,dt.$$

9.6 Uniform Convergence and Derivatives

We saw in Section 9.5 that a uniformly convergent sequence (or series) of continuous functions can be integrated term by term. This allows an easy proof of a theorem on term by term differentiation.

Theorem 9.34: *Let $\{f_n\}$ be a sequence of functions each with a continuous derivative on an interval $[a,b]$. If the sequence $\{f'_n\}$ of derivatives converges uniformly to a function on $[a,b]$ and the sequence $\{f_n\}$ converges pointwise to a function f, then f is differentiable on $[a,b]$ and*

$$f'(x) = \lim_n f'_n(x) \text{ for all } x \in [a,b].$$

Proof. Let $g = \lim_n f_n'$. Since each of the functions f_n' is assumed continuous and the convergence is uniform, the function g is also continuous (Theorem 9.22). From Theorem 9.29 we infer

$$\int_a^x g(t)\,dt = \lim_n \int_a^x f_n'(t)\,dt \text{ for all } x \in [a,b]. \tag{13}$$

Applying the fundamental theorem of calculus we see that

$$\int_a^x f_n'(t)\,dt = f_n(x) - f_n(a) \text{ for all } x \in [a,b] \tag{14}$$

for all $n \in \mathbb{N}$.

But $f_n(x) \to f(x)$ for all $x \in [a,b]$ by hypothesis, so letting $n \to \infty$ in equation (14) and noting (13), we obtain

$$\int_a^x g(t)\,dt = f(x) - f(a)$$

or

$$f(x) = \int_a^x g(t)\,dt + f(a).$$

It follows from the continuity of g and the fundamental theorem of calculus that f is differentiable and that

$$f'(x) = g(x)$$

for all $x \in [a,b]$. ■

For series, the theorem takes the following form:

Corollary 9.35: *Let $\{f_k\}$ be a sequence of functions each with a continuous derivative on $[a,b]$ and suppose $f = \sum_0^\infty f_k$ on $[a,b]$. If the series $\sum_{k=0}^\infty f_k'$ converges uniformly on $[a,b]$, then $f' = \sum_{k=0}^\infty f_k'$ on $[a,b]$.*

Example 9.36: Starting with the geometric series

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k \quad \text{on } (-1,1), \tag{15}$$

we obtain from Corollary 9.35 that

$$\frac{1}{(1-x)^2} = \sum_{k=1}^{\infty} kx^{k-1} \quad \text{on } (-1,1) \tag{16}$$

To justify (16) we observe first that the series (15) converges pointwise on $(-1,1)$. Next we note (Exercise 9.3.17) that the series (16) converges pointwise on $(-1,1)$ and uniformly on any closed interval $[a,b] \subset (-1,1)$. Thus, if $x \in (-1,1)$ and $-1 < a < x < b < 1$, then (16) converges uniformly on $[a,b]$, so (16) holds at x. ◄

9.6.1 Limits of Discontinuous Derivatives

✂ *Enrichment section. May be omitted.*

The hypotheses of Theorem 9.34 are somewhat more restrictive than necessary for the conclusion to hold. We need not assume that $\{f_n\}$ converges on all of $[a,b]$; convergence at a single point suffices. Nor need we assume that each of the derivatives f_n' is continuous. (We cannot, however, replace uniform convergence of the sequence $\{f'_n\}$

with pointwise convergence, as Example 9.5 shows.) The theorem that follows applies in a number of cases in which Theorem 9.34 does not apply.

Theorem 9.37: *Let $\{f_n\}$ be a sequence of continuous functions defined on an interval $[a,b]$. Suppose that $f_n'(x)$ exists for each n and each $x \in [a,b]$. Suppose that the sequence $\{f'_n\}$ of derivatives converges uniformly on $[a,b]$ and that there exists a point $x_0 \in [a,b]$ such that the sequence of numbers $\{f_n(x_0)\}$ converges. Then the sequence $\{f_n\}$ converges uniformly to a function f on the interval $[a,b]$, f is differentiable, and*

$$f'(x) = \lim_{n\to\infty} f_n'(x)$$

at each point $x \in [a,b]$.

Proof. Let $\varepsilon > 0$. Since the sequence of derivatives converges uniformly on $[a,b]$, there is an integer N_1 so that

$$|f_n'(x) - f_m'(x)| < \varepsilon$$

for all n, $m \geq N_1$ and all $x \in [a,b]$. Also, since the sequence of numbers $\{f_n(x_0)\}$ converges, there is an integer $N > N_1$ so that

$$|f_n(x_0) - f_m(x_0)| < \varepsilon$$

for all n, $m \geq N$. Let us, for any $x \in [a,b]$, $x \neq x_0$, apply the mean value theorem to the function $f_n - f_m$ on the interval $[x_0,x]$ (or on the interval $[x,x_0]$ if $x < x_0$). This gives us the existence of some point ξ between x and x_0 so that

$$f_n(x) - f_m(x) - [f_n(x_0) - f_m(x_0)] = (x - x_0)[f_n'(\xi) - f_m'(\xi)]. \tag{17}$$

From this we deduce that

$$\begin{aligned} |f_n(x) - f_m(x)| &\leq |f_n(x_0) - f_m(x_0)| + |(x - x_0)(f_n'(\xi) - f_m'(\xi)| \\ &< \varepsilon(1 + (b - a)) \end{aligned}$$

for any n, $m \geq N$. Since this N depends only on ε this assertion is true for all $x \in [a,b]$ and we have verified that the sequence of continuous functions $\{f_n\}$ is uniformly Cauchy on $[a,b]$ and hence converges uniformly to a continuous function f on $[a,b]$.

Let us now show that $f'(x_0)$ is the limit of the derivatives $f_n'(x_0)$. Again, for any $\varepsilon > 0$, equation (17) implies that

$$|f_n(x) - f_m(x) - [f_n(x_0) - f_m(x_0)]| \leq |x - x_0|\varepsilon \tag{18}$$

for all $n, m \geq N$ and any $x \neq x_0$ in the interval $[a,b]$. In this inequality let $m \to \infty$ and, remembering that $f_m(x) \to f(x)$ and $f_m(x_0) \to f(x_0)$, we obtain

$$|f_n(x) - f_n(x_0) - [f(x) - f(x_0)]| \leq |x - x_0|\varepsilon \tag{19}$$

if $n \geq N$. Let C be the limit of the sequence of numbers $\{f_n'(x_0)\}$. Thus there exists $M > N$ such that

$$|f_M'(x_0) - C| < \varepsilon. \tag{20}$$

Since the function f_M is differentiable at x_0, there exists $\delta > 0$ such that if $0 < |x - x_0| < \delta$, then

$$\left| \frac{f_M(x) - f_M(x_0)}{x - x_0} - f_M'(x_0) \right| < \varepsilon. \tag{21}$$

From Equation (19) and the fact that $M > N$, we have

$$\left| \frac{f_M(x) - f_M(x_0)}{x - x_0} - \frac{f(x) - f(x_0)}{x - x_0} \right| < \varepsilon.$$

This, together with the inequalities (20) and (21), shows that

$$\left| \frac{f(x) - f(x_0)}{x - x_0} - C \right| < 3\varepsilon$$

for $0 < |x - x_0| < \delta$. This proves that $f'(x_0)$ exists and is the number C, which we recall is $\lim_{n\to\infty} f_n'(x_0)$.

In this argument x_0 may be taken as any point inside the interval $[a,b]$ and so the theorem is proved. ■

For infinite series Theorem 9.37 takes the following form:

Corollary 9.38: *Let $\{f_k\}$ be a sequence of differentiable functions on an interval $[a,b]$. Suppose that the series $\sum_{k=0}^{\infty} f_k'$ converges uniformly on $[a,b]$. Suppose also that there exists $x_0 \in [a,b]$ such that the series $\sum_{k=0}^{\infty} f_k(x_0)$ converges. Then the series $\sum_{k=0}^{\infty} f_k(x)$ converges uniformly on $[a,b]$ to a function F, F is differentiable, and*

$$F'(x) = \sum_{k=0}^{\infty} f_k'(x)$$

for all $a \le x \le b$.

Note. In the statement of Theorem 9.37 we hypothesized the existence of a single point x_0 at which the sequence $\{f_n(x_0)\}$ converges. It then followed that the sequence $\{f_n\}$ converges on all of the interval I. If we drop that requirement but retain the requirement that the sequence $\{f_n'\}$ converges uniformly to a function g on I, we cannot conclude that $\{f_n\}$ converges on I [e.g. let $f_n(x) \equiv n$], but we can still conclude that there exists f such that $f' = g = \lim_n f_n'$ on I. (To see this, fix $x_0 \in I$, let $F_n = f_n - f_n(x_0)$ and apply Theorem 9.37 to the sequence $\{F_n\}$.) Thus, the uniform limit of a sequence of derivatives $\{f_n'\}$ is a derivative even if the sequence of primitives $\{f_n\}$ does not converge.

Exercises

9.6.1 Can the sequence of functions $f_n(x) = \dfrac{\sin nx}{n^3}$ be differentiated term by term? How about the series $\displaystyle\sum_{k=1}^{\infty} \frac{\sin kx}{k^3}$?

9.6.2 Verify that the function

$$y(x) = 1 + \frac{x^2}{1!} + \frac{x^4}{2!} + \frac{x^6}{3!} + \frac{x^8}{4!} + \dots$$

is a solution of the differential equation $y' = 2xy$ on $(-\infty, \infty)$ without first finding an explicit formula for $y(x)$.

9.7 Pompeiu's Function

✂ *Enrichment section. May be omitted.*

By the end of the nineteenth century analysts had developed enough tools to begin constructing examples of functions that challenged the then prevailing views. One famous mathematician, Henri Poincaré, complained that

> Before when one would invent a new function it was to some practical end; today they are invented to demonstrate the errors in the reasoning of our fathers ....

Many mathematicians were both shocked and appalled that functions could be constructed which possessed, to them, bizarre and unnatural properties. The beautiful and elegant theories of the nineteenth century were being torn to pieces by pathological examples.

Perhaps the earliest shock was the construction by Weierstrass and others of continuous functions that had derivatives at no points. This did indeed demonstrate some earlier errors because not a few mathematicians thought they had succeeded in proving that continuous functions could not be like this. Another famous example is due to Vito Volterra (1860–1940), who produced a differentiable function F with a bounded derivative F' that was not Riemann integrable.

In this section we present an example due to D. Pompeiu in 1906. This function h is differentiable and has the remarkable property that h' is discontinuous on a dense set and h' is zero on another dense set. We shall see that this implies that h is a differentiable function that, like Volterra's example, has a derivative that is not Riemann integrable. In fact, it is Riemann integrable on *no* interval while Volterra's example is Riemann integrable on many subintervals.

The example makes use of many theorems that we have established to this point and so offers an excellent review of our techniques. We present the example in a series of steps, each of which is left as a relatively easy exercise. (Exercise 9.7.4 is plausible but messy to verify, and you may prefer not to check the details.)

To begin the example we observe that the function

$$f(x) = \sqrt[3]{x-a}$$

has an infinite derivative at $x = a$ and a finite derivative elsewhere. Let $q_1, q_2, q_3, \ldots$ be an enumeration of $\mathbb{Q} \cap [0,1]$. Let

$$f(x) = \sum_{k=1}^{\infty} \frac{\sqrt[3]{x-q_k}}{10^k}.$$

The Pompeiu function is the inverse of this function, $h = f^{-1}$.

The details appear in the exercises. Note especially that our main goal is to prove that h is differentiable, h' is bounded, $h' = 0$ on a dense set and h' is positive and discontinuous on another dense set, and h' is not Riemann integrable.

For comparisons let us recall that in Exercise 7.4.2, we provided an example of a differentiable function g with g' bounded but discontinuous on a nowhere-dense perfect set P. Because of Section 8.6.3, we know that if P does not have measure zero, such a

function g' will not be Riemann integrable. Thus we cannot write

$$g(x) - g(a) = \int_a^x g'(t)\,dt,$$

that is, the fundamental theorem of calculus does not hold for the function g and its derivative g' if we are restricted to using the Riemann integral. This is essentially how Volterra constructed his example, by ensuring that the set P does not have measure zero.

We also mentioned in Section 7.4 that it is possible for a differentiable function f to have f' discontinuous on a dense set, and so Pompeiu's function justifies this comment.

Exercises

9.7.1 Show that the function $f(x) = (x-a)^{\frac{1}{3}}$ has an infinite derivative at $x = a$ and a finite derivative elsewhere.

9.7.2 Let $q_1, q_2, q_3, \ldots$ be an enumeration of $\mathbb{Q} \cap [0,1]$. For each $k \in \mathbb{N}$ let

$$f_k(x) = (x-q_k)^{\frac{1}{3}} \text{ and } f(x) = \sum_{k=1}^{\infty} \frac{f_k(x)}{10^k}.$$

Show that the series defining f converges uniformly.

9.7.3 Show that f is continuous on $[0,1]$.

9.7.4 Check that, for all $x \in \mathbb{R}$,

$$f'(x) = \sum_{k=1}^{\infty} \frac{f_k'(x)}{10^k} = \sum_{k=1}^{\infty} \frac{(x-q_k)^{-\frac{2}{3}}}{3 \times 10^k}.$$

(This part is messy to prove. Indicate why it is that we cannot simply apply Corollary 9.38 and differentiate term by term.)

9.7.5 Show that $f'(x) = \infty$ for all $x \in \mathbb{Q} \cap [0,1]$. (There are also other points at which f' is infinite; see Exercise 9.7.17.)

9.7.6 Show that $f([0,1])$ is an interval. Call it $[a,b]$.

9.7.7 Let $S = f(\mathbb{Q} \cap [0,1])$. Show that S is dense in $[a,b]$.

9.7.8 Show that f has an inverse.

9.7.9 Let $h = f^{-1}$. Show that h is continuous and strictly increasing on $[a,b]$.

9.7.10 Show that $h' = 0$ on the dense set S.

9.7.11 Show that there exists $\gamma > 0$ such that $f'(x) \geq \gamma$ for all $x \in [0,1]$.

9.7.12 Show that h is differentiable and that h' is bounded.

9.7.13 Show that $h' > 0$ on a dense subset of $[a,b]$.

9.7.14 Show that h' is discontinuous on a dense subset of $[a,b]$.

9.7.15 Thus far we know that h is differentiable, has a bounded derivative, $h' = 0$ on a dense set and h' is positive and discontinuous on another dense set. Show that h' is not Riemann integrable.

SEE NOTE 247

9.7.16 Show that $\{x : h'(x) \neq 0\}$ does not have measure zero.

9.7.17 Show that there exists $x \notin S$ such that $h'(x) = 0$ and that there exists $t \notin \mathbb{Q}$ such that $f'(t) = \infty$.

9.7.18 Show that the function h is not convex or concave in any interval. Which of the definitions of inflection point given as Exercise 7.10.14 apply to the points x at which $h'(x) = 0$? Do you think that such a point should be called an inflection point?

9.8 Continuity and Pointwise Limits

✂ *Advanced section. May be omitted.*

Much of this chapter focused on the concept of uniform convergence because of its role in providing affirmative answers to the questions we raised in Section 9.1. In particular, we saw in Section 9.2 that a pointwise limit of a sequence of continuous functions need not be continuous. On the other hand, these problems will not occur if the convergence is uniform.

There are, however, many situations in which pointwise convergence arises naturally, but uniform convergence doesn't. Consider, for example, a function F that is differentiable on $\mathbb{R}$. Then for $x \in \mathbb{R}$,

$$F'(x) = \lim_{n\to\infty} \frac{F(x+\frac{1}{n}) - F(x)}{\frac{1}{n}}.$$

If we define functions f_n by

$$f_n(x) = \frac{F(x+\frac{1}{n}) - F(x)}{\frac{1}{n}},$$

then each of the functions f_n is continuous on $\mathbb{R}$ and $f_n \to F'$ pointwise.

Now, we have seen examples of derivatives that are discontinuous at many points. For example, the function h' in Section 9.7 is discontinuous on a set that is dense in $[0,1]$ and does not have measure zero. Similarly, Exercise 7.4.2 provides an example of a differentiable function g whose derivative g' is discontinuous at every point of a Cantor set that does not have measure zero. We might ask the question, "Can the derivative of a differentiable function be discontinuous everywhere?" We shall see that the answer is "no." In fact, the set of points of continuity must be large in the sense of category—this set must be dense and of type $\mathcal{G}_\delta$, therefore residual (Theorem 6.17).

We actually prove a more general theorem.

Theorem 9.39: *Let $\{g_n\}$ be a sequence of continuous functions defined on an interval I and converging pointwise to a function g on I. Then the set of points of continuity of g forms a dense set of type $\mathcal{G}_\delta$ in I.*

Proof. Let us first outline the idea of the proof, leaving the formal proof for a moment. In Section 6.7 we defined the oscillation $\omega_f(x_0)$ of a function f at a point x_0 and showed (Theorem 6.25) that f is continuous at x_0 if and only if $\omega_f(x_0) = 0$. We now show that under the hypotheses of Theorem 9.39, $\omega_g(x)$ will be zero on a dense set. That will imply that g is continuous on a dense set. This set must be of type $\mathcal{G}_\delta$ (by Theorem 6.28).

We will argue by contradiction. We suppose that g is discontinuous at every point of some subinterval J. We will then use the Baire category theorem (Theorem 6.11) to show that there exists $n \in \mathbb{N}$ and an interval $H \subset J$ such that $\omega_g(x) \geq 1/n$ at every point of H. (This argument is valid for any function discontinuous at every point of an interval J.) We then use our hypotheses on g to show this is impossible. We do this by applying the Baire category theorem once again to obtain a subinterval K of H that g maps onto a set of diameter less than $1/n$. This implies that $\omega_f(x) < 1/n$ for every $x \in K$, a contradiction.

Now we can begin a formal proof of Theorem 9.39.

In order to obtain a contradiction, we suppose that g is discontinuous everywhere on some interval $J \subset I$. For each $n \in \mathbb{N}$, let

$$E_n = \{x \in J : \omega_g(x) \geq 1/n\}.$$

Each of the sets E_n is closed (by Theorem 6.27)and $J = \bigcup_{n=1}^{\infty} E_n$.

By the Baire category theorem there exists $n \in \mathbb{N}$ and an interval $H \subset J$ such that E_n is dense in H. The interval H has the property that g maps every subinterval of H onto a set of diameter at least $1/n$. We now show this not possible for g, a pointwise limit of continuous functions.

Let $\{I_k = (a_k, b_k)\}$ be a sequence of intervals, each of length less than $1/n$, such that

$$g(H) \subset \bigcup_{k=1}^{\infty} I_k.$$

For each k, let $H_k = g^{-1}(I_k) \cap H$. Then $H = \bigcup_{k=1}^{\infty} H_k$, but none of the sets H_k can contain an interval [since each H_k has length less than $1/n$, but $\omega_g(x) \geq 1/n$ for all $x \in H$].

Now

$$H_k = \{x : g(x) < b_k\} \cap \{x : g(x) > a_k\}.$$

By Exercise 9.8.4, each of these sets is of type $\mathcal{F}_\sigma$, thus $H_k = \bigcup_{j=1}^{\infty} H_{kj}$, with each of the sets H_{kj} closed. It follows that

$$H = \bigcup_{k=1}^{\infty} H_k = \bigcup_{k=1}^{\infty} \bigcup_{j=1}^{\infty} H_{kj}.$$

The interval H is expressed as a countable union of closed sets. It follows from the Baire category theorem that at least one of these sets, say H_{ij}, is dense in some interval $K \subset H$. Since H_{ij} is closed, $H_{ij} \supset K$. But this implies that $H_i \supset K$, which we have seen is not possible (since each of the sets H_k contains no intervals). This contradiction completes the proof. ■

Corollary 9.40: *Let f be differentiable on an interval (a,b). Then f' is continuous on a residual subset of (a,b). Thus the set of points of continuity of f' must be dense in (a,b).*

Note. Theorem 9.39 and Exercise 9.8.4 describe two important properties of functions that are pointwise limits of sequences of continuous functions. Each such function f is

continuous on a residual set, and every set of the form $\{x : f(x) > a\}$ or $\{x : f(x) < a\}$ is of type $\mathcal{F}_\sigma$.

Theorem 9.39 can be generalized. If P is a nonempty closed subset of the domain of f, then the function $f|P$ is continuous on a dense $\mathcal{G}_\delta$ subset of P.

The converses are also true[3]: A function f is a pointwise limit of a sequence of continuous functions on an interval I if and only if for every closed set $P \subset I$, f considered as a function defined on the set P is continuous on a dense $\mathcal{G}_\delta$ in P, and this happens if and only if every set of the form $\{x : f(x) > a\}$ or $\{x : f(x) < a\}$ is of type $\mathcal{F}_\sigma$.

These theorems have many applications. Functions that are pointwise limits of sequences of continuous functions are called *Baire* 1 functions. We have seen that this class of functions contains the class of derivatives. It also contains all monotonic functions and many other important classes of functions that arise in analysis.

The following exercises may be instructive. You may need to use one of the unproved statements in this section to work some of these exercises.

Exercises

9.8.1 Give an example of a function F that is differentiable on $\mathbb{R}$ such that the sequence

$$f_n(x) = n(F(x+1/n) - F(x)),$$

converges pointwise but not uniformly to F'.

SEE NOTE 248

9.8.2 Give an example of a function f that is Baire 1 and a real number a so that the sets $\{x : f(x) > a\}$ and $\{x : f(x) < a\}$ are not open. Show that, for your example, these sets are of type $\mathcal{F}_\sigma$.

9.8.3 Give an example of a function f that is Baire 1 and a real number a so that the sets $\{x : f(x) \geq a\}$ and $\{x : f(x) \leq a\}$ are not closed. Show that, for your example, these sets are of type $\mathcal{G}_\delta$.

9.8.4 Show that for any f that is Baire 1 and any real number a the sets

$$\{x : f(x) > a\} \quad \text{and} \quad \{x : f(x) < a\}$$

are of type $\mathcal{F}_\sigma$.

9.8.5 If f has only countably many discontinuities on an interval I, then f is a Baire 1 function. In particular, this is true for every monotonic function.

9.8.6 Let K be the Cantor set in $[0,1]$. Define

$$f(x) = \begin{cases} 1, & \text{if } x \in K \\ 0, & \text{elsewhere;} \end{cases}$$

and

$$g(x) = \begin{cases} 1, & \text{if } x \text{ is a one-sided limit point of } K \\ 0, & \text{elsewhere.} \end{cases}$$

(a) Show that f and g have exactly the same set of continuity points.

[3]Proofs of these statements can be found in I. P. Natanson, *Theory of Functions of a Real Variable*, Vol. II, Chapter XV, Fredrick Ungar Pub. Co., New York (1955) [English translation].

(b) Show that f is a Baire 1 function but g is not.

9.8.7 Let f be the characteristic function of the rationals. Show that f is not a Baire 1 function. Show that f is a pointwise limit of a sequence of Baire 1 functions. (Such functions are called functions of Baire class 2.)

9.9 Challenging Problems for Chapter 9

9.9.1 Let f_n be a sequence of functions converging pointwise to a function f on the interval $[0,1]$. Suppose that each function f_n is convex on $[0,1]$. Show that the convergence is uniform on any interval $[a,b] \subset (0,1)$. Need it be uniform on $[0,1]$?

9.9.2 Let $f_n : [0,1] \to \mathbb{R}$ be a sequence of continuous functions converging pointwise to a function f. If the convergence is uniform, prove that there is a finite number M so that $|f_n(x)| < M$ for all n and all $x \in [0,1]$. Even if the convergence is not uniform, show that there must be a subinterval $[a,b] \subset [0,1]$ and a finite number M so that $|f_n(x)| < M$ for all n and all $x \in [a,b]$.

SEE NOTE 249

9.9.3 Let E be a set of real numbers, fixed throughout this exercise. For any function f defined on E write

$$\|f\|_\infty = \sup_{x \in E} |f(x)|.$$

Show that

(a) $\|f\|_\infty = 0$ if and only if f is identically zero on E.

(b) $\|cf\|_\infty = |c|\|f\|_\infty$ for any real number c.

(c) $\|f+g\|_\infty \leq \|f\|_\infty + \|g\|_\infty$ for any functions f and g.

(d) $f_n \to f$ uniformly on E if and only if $\|f - f_n\|_\infty \to 0$ as $n \to \infty$.

(e) f_n converges uniformly on E if and only if $\|f_m - f_n\|_\infty \to 0$ as $n, m \to \infty$.

(f) Using $E = (0,1)$ and $f_n(x) = x^n$, compute $\|f_n\|_\infty$ and, hence, show that $\{f_n\}$ is not converging uniformly to zero on $(0,1)$.

Notes

[242]Exercise 9.2.7. The statements that are defined by inequalities (e.g., bounded, convex) or by equalities (e.g., constant, linear) will not lead to an interchange of two limit operations, and you should expect that they are likely true.

[243]Exercise 9.2.8. As the footnote to the exercise explains, this was Luzin's unfortunate attempt as a young student to understand limits. The professor began by saying "What you say is nonsense." He gave him the example of the double sequence $m/(m+n)$ where the limits as $m \to \infty$ and $n \to \infty$ cannot be interchanged and continued by insisting that "permuting two passages to the limit *must not be done*." He concluded with "Give it some thought; you won't get it immediately."

[244]Exercise 9.3.15. Use the Cauchy criterion for convergence of sequences of real numbers to obtain a candidate for the limit function f. Note that if $\{f_n\}$ is uniformly Cauchy on a set D, then for each $x \in D$, the sequence of real numbers $\{f_n(x)\}$ is a Cauchy sequence and hence convergent.

[245]Exercise 9.3.16.

$$S_n(x) = \sum_{k=0}^{n} x^k = \frac{1-x^n}{1-x}.$$

[246]Exercise 9.4.10. For part (b) consider

$$F_n(x) = f_n(x) + Mx$$

and apply Exercise 9.4.6.

[247]Exercise 9.7.15. Suppose h' were Riemann integrable. Explain why

$$h(x) - h(a) = \int_a^x h'(t)\,dt$$

for all $x \in [a,b]$. Now by considering an appropriate Riemann sum, since $h' = 0$ on a dense set, we would have

$$h(x) - h(a) = 0$$

for all $x \in [a,b]$. That should be a contradiction.

[248]Exercise 9.8.1. What properties would F' have to have if the convergence were uniform?

[249]Exercise 9.9.2. You will need to use the Baire category theorem for the second part of this.

Chapter 10

POWER SERIES

✂ If the material on lim sups and lim infs in Section 2.13 of Chapter 2 was omitted, that should be studied before attempting this chapter. The notion of a radius of convergence depends naturally on these concepts.

10.1 Introduction

One of the simplest and, arguably, the most important type of series of functions is the power series. This is a series of the form

$$\sum_0^\infty a_k x^k$$

or, more generally,

$$\sum_0^\infty a_k (x-c)^k.$$

It represents the notion of an "infinitely long" polynomial

$$a_0 + a_1 x + a_2 x^2 + \cdots + a_k x^k + \ldots.$$

The material we developed in Chapter 9 will allow us to show in this chapter that power series can be treated very much as if they were indeed polynomials in the sense that they can be integrated and differentiated term by term.

The main reason for developing this theory is that it allows a representation for functions as series. This enlarges considerably the class of functions that we can work with. Not all functions that arise in applications can be expressed as finite combinations of the elementary functions (i.e., as combinations of e^x, x^p, $\sin x$, $\cos x$, etc.). Thus, if we remain at the level of an elementary calculus class, we would be unable to solve many problems since we cannot express the functions needed for the solution in any way. For a large and important class of problems, the functions that can be represented as power series (the so-called analytic functions) are precisely the functions needed.

10.2 Power Series: Convergence

We begin with the formal definition of power series.

Definition 10.1: Let $\{a_k\}$ be a sequence of real numbers and let $c \in \mathbb{R}$. A series of the form

$$\sum_0^\infty a_k(x-c)^k = a_0 + a_1(x-c) + a_2(x-c)^2 + \dots$$

is called a *power series* centered at c. The numbers a_k are called the *coefficients* of the power series.

What can we say about the set of points on which the power series

$$\sum_0^\infty a_k(x-c)^k$$

converges? It is immediately clear that the series converges at its center c. What possibilities are there? A collection of examples illustrates the methods and also essentially all of the possibilities.

Example 10.2: The simple example

$$\sum_1^\infty k^k x^k = x + 4x^2 + 27x^3 + \dots$$

shows that a power series can diverge at every point other than its center. Observe that in this example $k^k x^k = (kx)^k$ does not approach 0 unless $x = 0$, so the series diverges for every $x \neq 0$ by the trivial test. Thus the set of convergence of this series is the set $\{0\}$. ◀

Example 10.3: The familiar geometric series

$$\sum_{k=0}^\infty x^k$$

should be considered the most elementary of all power series. We know that this series converges precisely on the interval $(-1,1)$ and diverges everywhere else. ◀

Example 10.4: The series

$$\sum_{k=1}^\infty \frac{x^k}{k}$$

has as coefficients $a_k = 1/k$ and the root test[1] supplies

$$s = \limsup_{k\to\infty} \sqrt[k]{|x|^k/k} = |x|.$$

(Verify this!) Thus the series converges on $(-1,1)$ and diverges for $|x| > 1$. At the two endpoints of the interval $(-1,1)$ a different test is required. We see that for $x = 1$ this is the familiar harmonic series and so diverges, while for $x = -1$ this is the familiar alternating harmonic series and so converges nonabsolutely. The interval of convergence is $[-1,1)$. Observe that the series converges at only one of the two endpoints of the interval. ◀

[1]The form of the root test needed to discuss power series uses the limit superior. For that the study of Section 2.13 may be required.

Example 10.5: The series

$$\sum_{k=1}^{\infty} \frac{x^k}{k^2}$$

converges on $[-1,1]$ and diverges otherwise. Again the root test (or the ratio test) is helpful here. Simpler, though, is to notice that

$$\left|\frac{x^k}{k^2}\right| \leq \frac{1}{k^2}$$

for all $|x| \leq 1$ and so obtain convergence on $[-1,1]$ by a comparison test with the convergent series $\sum_{k=0}^{\infty} 1/k^2$. If $|x| > 1$ the terms $|x^k/k^2| \to \infty$ and so, trivially, the series diverges. Note here that the series converges on the interval $[-1,1]$ and is absolutely convergent there. ◀

Example 10.6: The root test applied to the series

$$\sum_{k=1}^{\infty} \frac{x^k}{k^k}$$

gives

$$\lim_{k\to\infty} \sqrt[k]{\frac{|x|^k}{k^k}} = |x| \lim_{k\to\infty} \frac{1}{k} = 0.$$

(The ratio test can also be used here.) It follows that the series converges for all $x \in \mathbb{R}$. Perhaps an easier method in this particular example is to use the comparison test and the fact that

$$\left|\frac{x}{k}\right|^k < \frac{1}{2^k} \text{ when } k \geq 2|x|.$$

Thus the series converges at any x by comparison with a geometric series. Thus the set of convergence of this series is $(-\infty,\infty)$, again as in the previous examples an interval. ◀

In general, as these examples seem to suggest, the set of points of convergence of a power series forms an interval and an application of the root test is an essential tool in determining that interval. Let us apply this test to the series

$$\sum_{0}^{\infty} a_k (x-c)^k.$$

Let

$$s = \limsup_{k\to\infty} \sqrt[k]{|a_k|}.$$

Then

$$\limsup_{k\to\infty} \sqrt[k]{|a_k||x-c|^k} = \limsup_{k\to\infty} \sqrt[k]{|a_k|}|x-c| = s|x-c|.$$

By the root test the series converges absolutely if $s|x-c| < 1$ and diverges if $s|x-c| > 1$.

If $0 < s < \infty$, then the series converges on the interval

$$(c - 1/s, c + 1/s)$$

and diverges for x outside the interval

$$[c - 1/s, c + 1/s].$$

The root test is inconclusive about convergence at the endpoints

$$x = c \pm 1/s$$

of these intervals. The interval of convergence is thus one of the four possibilities

$$(c-1/s, c+1/s) \text{ or } [c-1/s, c+1/s) \text{ or}$$

$$(c-1/s, c+1/s] \text{ or } [c-1/s, c+1/s].$$

If $s = 0$, then the series converges for all values of x. We could say that the interval of convergence is $(-\infty, \infty)$ in this case. If $s = \infty$, then the series converges for no values of x other than the trivial value $x = c$. We could say that the interval of convergence is the degenerate "interval" $\{c\}$.

Thus the set of convergence is an interval centered at c. This interval might be degenerate (consisting of only the center), might be all of the real line, and might contain none, one, or both of its endpoints.

Our next theorem summarizes the discussion of convergence to this point. We first give a formal definition.

Definition 10.7: Let $\sum_0^\infty a_k(x-c)^k$ be a power series. Then the number

$$R = \frac{1}{\limsup_{k\to\infty} \sqrt[k]{|a_k|}}$$

is called the *radius of convergence* of the series. Here we interpret $R = \infty$ if

$$\limsup_{k\to\infty} \sqrt[k]{|a_k|} = 0$$

and $R = 0$ if

$$\limsup_{k\to\infty} \sqrt[k]{|a_k|} = \infty.$$

Note. This book deals with *real* analysis, but a full theory of power series fits more naturally into the setting of *complex* analysis. In that setting, a power series converges in a "circle of convergence" centered at a complex number c in the complex plane and with radius

$$R = \frac{1}{\limsup_k \sqrt[k]{|a_k|}}.$$

This explains the origin of the term "radius of convergence."

Theorem 10.8: *Let $\sum_0^\infty a_k(x-c)^k$ be a power series with radius of convergence R.*

1. *If $R = 0$, then the series converges only at $x = c$.*
2. *If $R = \infty$, then the series converges absolutely for all x.*
3. *If $0 < R < \infty$, then the series converges absolutely for all x satisfying $|x-c| < R$ and diverges for all x satisfying $|x-c| > R$.*

Proof. We first consider the case $R = 0$. Here $\limsup_k \sqrt[k]{|a_k|} = \infty$ so, for $x \neq c$,

$$\limsup_k \sqrt[k]{|a_k||x-c|^k} = |x-c| \limsup_k \sqrt[k]{|a_k|} = \infty.$$

By the root test the series cannot converge for $x \neq c$. The other cases are similarly established by the root test as in the discussion following our examples. ■

In general, a power series

$$\sum_{k=0}^{\infty} a_k x^k$$

with a finite radius of convergence R must have as its set of convergence one of the four intervals

$$(-R,R), \quad [-R,R], \quad (-R,R] \text{ or } \quad [-R,R).$$

As we have seen from the examples, each of these four cases can occur. The other possibilities are for series with radius of convergence $R = 0$, in which case the set of convergence is trivially $\{0\}$, or with radius of convergence $R = \infty$, in which case the set of convergence is the entire real line. Note too that if the series converges absolutely at $x = R$ or at $x = -R$, then it must converge absolutely on all of $[-R,R]$. It is possible, though, for the series to converge nonabsolutely at one endpoint but not at the other.

Exercises

10.2.1 Find the radius of convergence for each of the following series.

(a) $\displaystyle\sum_{k=0}^{\infty} (-1)^k x^{2k}$

(b) $\displaystyle\sum_{k=0}^{\infty} \frac{x^k}{k!}$

(c) $\displaystyle\sum_{k=0}^{\infty} k x^k$

(d) $\displaystyle\sum_{k=0}^{\infty} k! x^k$

10.2.2 If the limit

$$\lim_{k\to\infty} \left| \frac{a_k}{a_{k+1}} \right|$$

exists or equals ∞, then show that the following expression also gives the radius of convergence of a power series:

$$R = \lim_{k\to\infty} \left| \frac{a_k}{a_{k+1}} \right|.$$

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10.2.3 For the examples

$$\sum_{k=0}^{\infty} x^k, \quad \sum_{k=1}^{\infty} \frac{x^k}{k}, \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{x^k}{k^2}$$

verify in each case that

$$R = \lim_k \left| \frac{a_k}{a_{k+1}} \right| = 1.$$

10.2.4 For the series

$$\sum_{k=1}^{\infty} k^k x^k \quad \text{and} \quad \sum_{k=1}^{\infty} \frac{x^k}{k^k}$$

check that the radius of convergence is $R = 0$ and ∞, respectively.

10.2.5 Give an example of a power series $\sum_0^\infty a_k x^k$ for which the radius of convergence R satisfies

$$R = \frac{1}{\lim_{k\to\infty} \sqrt[k]{|a_k|}}$$

but

$$\lim_{k\to\infty} \left| \frac{a_k}{a_{k+1}} \right|$$

does not exist.

10.2.6 Give an example of a power series $\sum_0^\infty a_k x^k$ for which the radius of convergence R satisfies

$$\liminf_k \left| \frac{a_{k+1}}{a_k} \right| < R < \limsup_k \left| \frac{a_{k+1}}{a_k} \right|.$$

10.2.7 Give an example of a power series $\sum_0^\infty a_k x^k$ with radius of convergence 1 that is non-absolutely convergent at both endpoints 1 and -1 of the interval of convergence.

10.2.8 Give an example of a power series $\sum_0^\infty a_k x^k$ with interval of convergence exactly $[-\sqrt{2}, \sqrt{2})$.

10.2.9 If the power series $\sum_0^\infty a_k x^k$ has a radius of convergence R, what must be the radius of convergence of the series

$$\sum_{k=0}^{\infty} k a_k x^k \text{ and } \sum_{k=1}^{\infty} k^{-1} a_k x^k?$$

10.2.10 If the coefficients $\{a_k\}$ of a power series $\sum_0^\infty a_k x^k$ form a bounded sequence show that the radius of convergence is at least 1.

10.2.11 If the power series $\sum_0^\infty a_k x^k$ has a radius of convergence R_a and the power series $\sum_0^\infty b_k x^k$ has a radius of convergence R_b and $|a_k| \le |b_k|$ for all k sufficiently large, what relation must hold between R_a and R_b?

10.2.12 If the power series $\sum_0^\infty a_k x^k$ has a radius of convergence R, what must be the radius of convergence of the series $\sum_{k=0}^\infty a_k x^{2k}$?

10.2.13 If the power series $\sum_0^\infty a_k x^k$ has a finite positive radius of convergence show that the radius of convergence of the series $\sum_{k=0}^\infty a_k x^{k^2}$ is 1.

10.2.14 Find the radius of convergence of the series

$$\sum_{k=0}^{\infty} \frac{(\alpha k)!}{(k!)^\beta} x^k,$$

where α and β are positive and α is an integer.

10.2.15 Let $\{a_k\}$ be a sequence of real numbers and let $x_0 \in \mathbb{R}$. Suppose there exists $M > 0$ such that $|a_k x_0^k| \le M$ for all $k \in \mathbb{N}$. Prove that $\sum_0^\infty a_k x^k$ converges absolutely for all x satisfying the inequality $|x| < |x_0|$. What can you say about the radius of convergence of this series?

10.3 Uniform Convergence

So far we have reached a complete understanding of the nature of the set of convergence of any power series. In order to apply many of our theorems of Chapter 9 to questions concerning term by term integration or differentiation of power series, we need to check questions related to the *uniform* convergence of power series. Our next theorem does this and also summarizes the discussion of convergence to this point.

We repeat the convergence results of Theorem 10.8 but now add a discussion of uniform convergence.

Theorem 10.9: *Let $\sum_0^\infty a_k(x-c)^k$ be a power series with radius of convergence R.*

1. *If $R = 0$, then the series converges only at $x = c$.*
2. *If $R = \infty$, then the series converges absolutely and uniformly on any compact interval $[a,b]$.*
3. *If $0 < R < \infty$, then the series converges absolutely and uniformly on any interval $[a,b]$ contained entirely inside the interval $(c-R, c+R)$.*

Proof. To verify (2) and (3), let us choose $0 < \rho < R$ so that the interval $[a,b]$ is contained inside the interval $(c-\rho, c+\rho)$. Fix $\rho_0 \in (\rho, R)$. Then

$$\limsup_k \sqrt[k]{|a_k|} = \frac{1}{R} < \frac{1}{\rho_0}.$$

Thus there exists $N \in \mathbb{N}$ such that

$$\sqrt[k]{|a_k|} < \frac{1}{\rho_0} \quad \text{for all } k \geq N. \tag{1}$$

For $k \geq N$ and $|x-c| \leq \rho$ we calculate

$$|a_k(x-c)^k| \leq |a_k|\rho^k < \left(\frac{\rho}{\rho_0}\right)^k,$$

the last inequality following from (1).

Now since $\rho/\rho_0 < 1$, it follows that

$$\sum_{k=0}^{\infty} \left(\frac{\rho}{\rho_0}\right)^k < \infty.$$

It now follows from the Weierstrass M-test (Theorem 9.16) that the series converges absolutely and uniformly on the set $\{x : |x-c| < \rho\}$ and hence also on the subset $[a,b]$. ■

If the interval of convergence of a power series is $(-R,R)$, then certainly the assertion (3) of Theorem 10.9 is the best that can be made. (See Exercise 10.3.3.) The geometric series $\sum_{n=0}^\infty x^n$ furnishes the clearest example of this. This series converges on $(-1,1)$ but does not converge uniformly on the entire interval of convergence $(-1,1)$. It does, however, converge uniformly on any $[a,b] \subset (-1,1)$.

To improve on this we can ask the following: If R is the radius of convergence of a power series and the interval of convergence is $[-R,R]$ or $(-R,R]$ or $[-R,R)$, can

uniform convergence be extended to the endpoints? If the convergence at an endpoint R (or $-R$) is absolute, then an application of the Weierstrass M-test shows immediately that the convergence is absolute and uniform on $[-R,R]$. For nonabsolute convergence a more delicate test is needed and we need to appeal to material developed in Section 9.3.3. The following theorem contains, for easy reference, a repeat of the third assertion in Theorem 10.9.

Theorem 10.10: *Suppose that the power series $\sum_0^\infty a_k(x-c)^k$ has a finite and positive radius of convergence R and an interval of convergence I.*

1. *If $I=[c-R,c+R]$, then the series converges uniformly (but not necessarily absolutely) on $[c-R,c+R]$.*
2. *If $I=(c-R,c+R]$, then the series converges uniformly (but not necessarily absolutely) on any interval $[a,c+R]$ for all*
$$c-R<a<c+R.$$
3. *If $I=[c-R,c+R)$, then the series converges uniformly (but not necessarily absolutely) on any interval $[c-R,b]$ for all*
$$c-R<b<c+R.$$
4. *If $I=(c-R,c+R)$, then the series converges uniformly and absolutely on any interval $[a,b]$ for $c-R<a<b<c+R$.*

Proof. For the purposes of the proof we can take $c=0$. Let us examine the case
$$I=(c-R,c+R]=(-R,R]$$
which is typical. Consider the intervals $[a,0]$ for $-R<a<0$ and $[0,R]$. The uniform convergence of the series on $[a,0]$ is clear since this is contained entirely inside the interval of convergence.

Now we examine uniform convergence on $[0,R]$. We consider the series
$$\sum_{k=0}^{\infty} a_k x^k = \sum_{k=0}^{\infty} A_k t^k,$$
where $A_k=a_kR^k$ and $t=(x/R)$. The series $\sum_{k=0}^\infty A_kt^k$ converges for $0\le t\le 1$ by our assumptions. Note that $\sum_{k=0}^\infty A_k$ is convergent while the sequence $\{t^k\}$ converges monotonically on the interval $[0,1]$. By a variant of Theorem 9.19 (Exercise 9.3.26) this series converges uniformly for t in the interval $[0,1]$. This translates easily to the assertion that our original series converges uniformly for $x\in[0,R]$. Thus since the series converges uniformly on $[a,0]$ and on $[0,R]$ we have obtained the uniform convergence on $[a,R]$ as required. The other cases are similarly handled. ■

Exercises

10.3.1 Characterize those power series $\sum_0^\infty a_k(x-c)^k$ that converge uniformly on $(-\infty,\infty)$.

10.3.2 Show that if $\sum_{k=0}^\infty a_kx^k$ converges absolutely at a point $x_0>0$, then the convergence of the series is uniform on $[-x_0,x_0]$.

10.3.3 Show that if $\sum_{k=0}^{\infty} a_k x^k$ converges uniformly on an interval $(-r,r)$, then it must in fact converge uniformly on $[-r,r]$. Deduce that if the interval of convergence is exactly of the form $(-R,R)$, or $[-R,R)$ or $[-R,R)$, then the series cannot converge uniformly on the entire interval of convergence.
SEE NOTE 251

10.4 Functions Represented by Power Series

Suppose now that a power series $\sum_0^\infty a_k(x-c)^k$ has positive or infinite radius of convergence R. Then this series represents a function f on (at least) the interval $(c-R,c+R)$:

$$f(x)=\sum_0^\infty a_k(x-c)^k \quad \text{for } |x-c|<R. \tag{2}$$

If the series converges at one or both endpoints, then this represents a function on $[c-R,c+R)$ or $(c-R,c+R]$ or $[c-R,c+R]$.

What can we say about the function f? In terms of the questions that have motivated us throughout Chapter 9 we can ask

1. Is the function f continuous on its domain of definition?
2. Can f be differentiated by termwise differentiation of its series?
3. Can f be integrated by termwise integration of its series?

We address each of these questions and find that generally the answer to each is yes.

10.4.1 Continuity of Power Series

A power series may represent a function on an interval. Is that function necessarily continuous?

Theorem 10.11: *A function f represented by a power series*

$$f(x)=\sum_0^\infty a_k(x-c)^k \tag{3}$$

is continuous on its interval of convergence.

Proof. This follows from Theorem 10.10. For example, if the interval of convergence is $(c-R,c+R]$, then we can show that f is continuous at each point of this interval. Since convergence is uniform on $[c,c+R]$ and since each of the functions $a_k(x-c)^k$ is continuous on $[c,c+R]$, the same is true of the function f (Corollary 9.23). For any point $x_0\in(c-R,c)$ we can similarly prove that f is continuous at x_0 in the same way by noting that the series converges uniformly on an interval $[a,c]$, where a is chosen so that $c-R<a<x_0<c$. ■

Example 10.12: The series

$$f(x)=\sum_{k=1}^\infty \frac{x^k}{k}$$

converges at every point of the interval $[-1,1)$. Consequently, this function is continuous at every point of that interval. We shall show in the next section that the identity

$$\log(1-x) = \sum_{k=1}^{\infty} \frac{x^k}{k}$$

holds for all $x \in (-1,1)$ (by integrating the geometric series term by term). Since we are also assured of continuity at the endpoint $x = -1$ we can conclude that

$$\log 2 = \sum_{k=1}^{\infty} \frac{(-1)^k}{k}.$$

◀

10.4.2 Integration of Power Series

If a function is represented by a power series, is it possible to integrate that function by integrating the power series term by term?

Theorem 10.13: *Let a function f be represented by a power series*

$$f(x) = \sum_{0}^{\infty} a_k (x-c)^k$$

with an interval of convergence I. Then for every point x in that interval f is integrable on $[c,x]$ (or $[x,c]$ if $x < c$) and

$$\int_c^x f(t)\,dt = \sum_{k=0}^{\infty} \frac{a_k}{k+1}(x-c)^{k+1}.$$

Proof. Let x be a point in the interval of convergence. The convergence of the series $\sum_0^\infty a_k(x-c)^k$ is uniform on $[c,x]$ (or on $[x,c]$ if $x < c$), so the series can be integrated term by term (Theorem 9.29). ■

Example 10.14: The geometric series

$$\frac{1}{1-x} = \sum_{k=0}^{\infty} x^k$$

has radius of convergence 1 and so can be integrated term by term provided we stay inside the interval $(-1,1)$. Thus

$$-\log(1-x) = \int_0^x \frac{1}{1-t}\,dt = \sum_{k=0}^{\infty} \frac{1}{k+1} x^{k+1}$$

for all $-1 < x < 1$. We would not be able to conclude from this theorem that the integral can be extended to the endpoints of $(-1,1)$. The new series, however, also converges at $x = -1$ and so we can apply Theorem 10.11 to show that the identity just proved is actually valid on $[-1,1)$. ◀

10.4.3 Differentiation of Power Series

If a function is represented by a power series, is it possible to differentiate that function by differentiating the power series term by term?

Note that for continuity and integration we were able to prove Theorems 10.11 and 10.13 immediately from general theorems on uniform convergence. To prove a theorem on term by term differentiation, we need to check uniform convergence of the series of *derivatives*. The following lemma gives us what we need.

Lemma 10.15: *Let $\sum_0^\infty a_k(x-c)^k$ have radius of convergence R. Then the series*

$$\sum_{k=1}^{\infty} ka_k(x-c)^{k-1}$$

obtained via term by term differentiation also has the same radius of convergence R.

Proof. The radius of convergence of the series is given by

$$R = \frac{1}{\limsup_k \sqrt[k]{|a_k|}}.$$

The radius of convergence of the differentiated series is given by

$$R' = \frac{1}{\limsup_k \sqrt[k]{|ka_k|}}.$$

But since $\sqrt[k]{k} \to 1$ as $k \to \infty$ we see immediately that these two expressions are equal. (They may be both zero or both infinite.) ■

Theorem 10.16: *Let $\sum_0^\infty a_k(x-c)^k$ have radius of convergence $R > 0$, and let*

$$f(x) = \sum_0^{\infty} a_k(x-c)^k$$

whenever $|x-c| < R$. Then f is differentiable on $(c-R, c+R)$ and

$$f'(x) = \sum_{k=1}^{\infty} ka_k(x-c)^{k-1}$$

for each $x \in (c-R, c+R)$.

Proof. It follows from the preceding lemma that the series

$$\sum_{k=1}^{\infty} ka_k(x-c)^{k-1}$$

has radius of convergence R. Thus this series converges uniformly on any compact interval $[a,b]$ contained in $(c-R, c+R)$. Since each value of x in $(c-R, c+R)$ can be placed inside some such interval $[a,b]$ it now follows immediately from Corollary 9.35 that $f'(x) = \sum_{k=1}^\infty ka_k(x-c)^{k-1}$ whenever $|x-c| < R$. ■

We can apply the same argument to the differentiated series and differentiate once more. From the expansion

$$f'(x) = \sum_{k=1}^{\infty} ka_k(x-c)^{k-1}$$

we obtain a formula for $f''(x)$:

$$f''(x) = \sum_{2}^{\infty} k(k-1)a_k(x-c)^{k-2}.$$

Let us express explicitly the formulas of $f(x)$, $f'(x)$, and $f''(x)$.

$$\begin{aligned} f(x) &= a_0 + a_1(x-c) + a_2(x-c)^2 + a_3(x-c)^3 + \dots \\ f'(x) &= a_1 + 2a_2(x-c) + 3a_3(x-c)^2 + \dots \\ f''(x) &= 2a_2 + 3\cdot 2a_3(x-c) + \dots \end{aligned}$$

These expressions are valid in the interval $(c-R, c+R)$. For $x = c$ we obtain

$$\begin{aligned} f(c) &= a_0 \\ f'(c) &= a_1 \\ f''(c) &= 2a_2. \end{aligned}$$

If we continue in this way, we can obtain power series expansions for all the derivatives of f. This results in the following theorem. The proof (which requires mathematical induction) is left as Exercise 10.4.1.

Theorem 10.17: *Let $\sum_0^\infty a_k(x-c)^k$ have radius of convergence $R > 0$. Then the function*

$$f(x) = \sum_{0}^{\infty} a_k(x-c)^k$$

has derivatives of all orders and these derivatives can be calculated by repeated term by term differentiation. The coefficients a_k are related to the derivatives of f at $x = c$ by the formula

$$a_k = \frac{f^{(k)}(c)}{k!}.$$

Uniqueness of Power Series From Theorem 10.17 we deduce that any two power series representations of a function must be identical. Note that the centers have to be the same for this to be true.

Corollary 10.18: *Suppose two power series*

$$f(x) = \sum_{0}^{\infty} a_k(x-c)^k$$

and

$$g(x) = \sum_{0}^{\infty} b_k(x-c)^k$$

agree on some interval centered at c, that is $f(x) = g(x)$ for $|x-c| < \rho$ and some positive ρ. Then $a_k = b_k$ for all $k = 0, 1, 2, \dots$.

Proof. It follows immediately from Theorem 10.17 that

$$a_k = \frac{f^{(k)}(c)}{k!} = \frac{g^{(k)}(c)}{k!} = b_k$$

for all $k = 0, 1, 2, \dots$. ■

Example 10.19: The series for the exponential function

$$e^x = \sum_{k=0}^{\infty} \frac{x^k}{k!}$$

reveals one of the key facts about the exponential function, namely that it is its own derivative. Note simply that

$$\frac{d}{dx}e^x = \frac{d}{dx}\sum_{k=0}^{\infty}\frac{x^k}{k!} = \sum_{k=0}^{\infty}\frac{d}{dx}\frac{x^k}{k!} = \sum_{k=1}^{\infty}\frac{x^{k-1}}{k-1!} = e^x.$$

◀

Example 10.20: The material in this section can also be used to obtain the power series expansion of the exponential function. Suppose that we know that the exponential function $f(x) = e^x$ does in fact have a power series expansion

$$f(x) = \sum_{k=0}^{\infty} a_k x^k.$$

Then the coefficients must be given by the formulas we have obtained, namely

$$a_k = \frac{f^{(k)}(0)}{k!}.$$

But for $f(x) = e^x$ it is clear that $f^{(k)}(0) = 1$ for all k and so the series must be indeed be given by $a_k = 1/k!$ as we well know. But how can we be assured that the exponential function does have a power series expansion? This argument shows only that if there is a series, then that series is precisely $\sum_{k=0}^{\infty}\frac{x^k}{k!}$. There remains the possibility that there may not be a series after all. This is the subject of the next section. ◀

10.4.4 Power Series Representations

Corollary 10.18 shows that if we can obtain a power series representation for a function f by any means whatsoever, then that series must have its coefficients given by the equations $a_k = f^{(k)}(c)/k!$. In particular, a power series representation for f about a given point must be unique.

Example 10.21: For example, the formula for the sum of a geometric series can be used to show that

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots + (-1)^j x^{2j} + \ldots.$$

Thus this series represents the function $f(x) = \frac{1}{1+x^2}$ on the interval $(-1,1)$. Note that the coefficients a_k are zero if k is odd and that $a_{2j} = (-1)^j$ for $k = 2j$ even. It now follows *automatically* that for even integers $k = 2j$

$$\frac{f^{(k)}(0)}{k!} = a_k = (-1)^j$$

while all the odd derivatives are zero. Thus

$$\frac{d^k}{dx^k}\left(\frac{1}{1+x^2}\right) = 0 \quad \text{at } x = 0$$

if k is odd and, if $k = 2j$ is even,

$$\frac{d^k}{dx^k}\left(\frac{1}{1+x^2}\right) = (-1)^j(2j)! \quad \text{at } x = 0.$$

◀

Note. There is a curious fact here that should be puzzled upon. The formula

$$\frac{1}{1+x^2} = 1 - x^2 + x^4 - \cdots + (-1)^j x^{2j} + \ldots$$

is valid precisely for $-1 < x < 1$. But the function on the left-hand side of this identity is defined for all values of x. We might have hoped for a representation valid for all x, but we do not obtain one!

Sometimes the easiest way to obtain a power series expansion formula for a function is by using the formula

$$a_k = \frac{f^{(k)}(c)}{k!}.$$

For example, this is how we obtained the power series for $f(x) = e^x$. We compute $f^{(k)}(x) = e^x$ for $k = 0, 1, 2, \ldots$, so $f^{(k)}(0) = 1$ for all k. Thus the series expansion for this function (if it has a series expansion) would have to be

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots = \sum_0^\infty \frac{x^k}{k!}. \tag{4}$$

Note that the series converges for all $x \in \mathbb{R}$. In the next section we will show how to verify that the equality holds for all x.

If we had wanted a formula for $g(x) = e^{x^2}$ we might have used the same idea and determined all the derivatives $g^{(k)}(0)$. It would be simplest, however, to just substitute x^2 for x in the expansion (4), obtaining

$$e^{x^2} = 1 + x^2 + \frac{x^4}{2!} + \frac{x^6}{3!} + \cdots = \sum_0^\infty \frac{x^{2k}}{k!}. \tag{5}$$

Also, from this expansion we can readily obtain an expansion for $2xe^{x^2}$ in either of two ways: We can multiply the expansion in (5) by $2x$ giving

$$2xe^{x^2} = 2x + 2x^3 + \frac{2x^5}{2!} + \frac{2x^7}{3!} + \cdots = \sum_0^\infty \frac{2x^{2k+1}}{k!}.$$

Alternatively, we can use Theorem 10.16 and differentiate (5) term by term, giving

$$2xe^{x^2} = \frac{d}{dx}e^{x^2} = 2x + \frac{4x^3}{2!} + \frac{6x^5}{3!} + \frac{8x^7}{4!} \cdots = \sum_0^\infty \frac{2x^{2k+1}}{k!}.$$

You may wish instead to try to obtain these expansions directly by using the formula $a_k = f^{(k)}(c)/k!$.

Exercises

10.4.1 Provide the details in the proof of Theorem 10.17.

10.4.2 Obtain expansions for

$$\frac{x}{1+x^2} \text{ and } \frac{x}{(1+x^2)^2}.$$

10.4.3 Obtain expansions for

$$\frac{1}{1+x^3} \text{ and } \frac{x^2}{1+x^3}.$$

10.4.4 Find a power series expansion about $x = 0$ for the function

$$f(x) = \int_0^1 \frac{1-e^{-sx}}{s}\,ds.$$

SEE NOTE 252

10.4.5 The function

$$J_0(x) = \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k}}{(k!)^2 2^{2k}}$$

is called a Bessel function of order zero of the first kind. Show that this is defined for all values of x. The function $J_1(x) = -J_0'(x)$ is called a Bessel function of order one of the first kind. Find a series expansion for $J_1(x)$.

10.4.6 Let

$$f(x) = \sum_{k=0}^{\infty} a_k x^k$$

have a positive radius of convergence. If the function f is *even* [i.e., if it satisfies $f(-x) = f(x)$ for all x], what can you deduce about the coefficients a_k? What can you deduce if the function is *odd* (i.e., if $f(-x) = -f(x)$ for all x)?

10.4.7 Let

$$f(x) = \sum_{k=0}^{\infty} a_k x^k$$

have a positive radius of convergence. If zero is a critical point (i.e., if $a_1 = 0$) and if $a_2 > 0$, then the point $x = 0$ is a strict local minimum. Prove this and also formulate and prove a generalization of this that would allow

$$a_2 = a_3 = a_4 = \cdots = a_{N-1} = 0 \text{ and } a_N \neq 0.$$

10.5 The Taylor Series

We have seen that if a power series $\sum_0^\infty a_k (x-c)^k$ converges on an interval I, then the series represents a function f that has derivatives of all orders. In particular, the coefficients a_k are related to the derivatives of f at c:

$$a_k = \frac{f^{(k)}(c)}{k!}.$$

We then call the series the *Taylor series* for f about the point $x = c$.

Let us turn the question around:

What functions f have a Taylor series representation in their domain?

We see immediately that such a function must be infinitely differentiable in a neighborhood of c since for such a series to be valid we know that all of the derivatives $f^{(k)}(c)$ must exist. But is that enough?

If we start with a function that has derivatives of all orders on an interval I containing the point c and write the series

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}(x-c)^k,$$

we might expect that this is exactly the representation we want. Indeed *if there is a valid representation*, then this must be the one, since such representations are unique. But can we be sure the series converges to f on I? Or even that the series converges at all on I. The answer to both questions is "no."

Example 10.22: Consider, for example, the function

$$f(x) = 1/(1+x^2).$$

This function is infinitely differentiable on all of the real line. Its Taylor series about $x = 0$ is, as we have seen in Example 10.21,

$$1 - x^2 + x^4 - x^6 + \cdots = \sum_{k=0}^{\infty} (-1)^k x^{2k}.$$

This series converges for $|x| < 1$ but diverges for $|x| \geq 1$. It does represent f on the interval $(-1,1)$ but not on the full domain of f. Indeed there can be no series of the form $\sum_{k=0}^{\infty} a_k x^k$ that represents f on $(-\infty,\infty)$ since that series would agree with this present series on $(-1,1)$ and so could not be any different. ◀

Worse situations are possible. For example, there are infinitely differentiable functions whose Taylor series have zero radius of convergence for every c; for these functions

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(c)}{k!}(x-c)^k$$

diverges except at $x = c$ and this is true for all $c \in \mathbb{R}$.[2] For these functions the Taylor series cannot represent the function.

Another unpleasant situation occurs when a Taylor series converges to the wrong function. This possibility seems even more startling!

Example 10.23: Consider the function

$$f(x) = \begin{cases} 0, & \text{if } x = 0 \\ e^{-1/x^2}, & \text{if } x \neq 0. \end{cases}$$

Exercise 10.5.4 provides an outline for showing that f is infinitely differentiable on the real line, and that $f^{(k)}(0) = 0$ for $k = 1,2,3,\ldots$. Thus the Taylor series for f about $x = 0$ takes the form $\sum_{k=0}^{\infty} 0x^k$ with all coefficients equal to zero. This series converges to the zero function on the real line, so it does not represent f except at the origin, even though the series converges for all x. ◀

[2] See D. Morgenstern, *Math. Nach.*, **12** (1954), p. 74. We find here that in a certain sense "most" infinitely differentiable functions have this property!

10.5.1 Representing a Function by a Taylor Series

The preceding discussion shows that we should not automatically assume that a Taylor series for a function f represents f. It is true, however, that the developments in the earlier sections of this chapter help us see that many of the familiar Taylor series encountered in elementary calculus are valid.

Example 10.24: For example, starting with the geometric series

$$\frac{1}{1+x} = \sum_{k=0}^{\infty} (-1)^k x^k,$$

we can apply Theorem 10.13 on integrating a power series term by term to obtain, for $|x| < 1$,

$$\begin{aligned} \ln(1+x) &= \int_1^x \frac{1}{1+t}\,dt = \sum_{k=0}^{\infty} \int_0^x (-1)^k t^k\,dt \\ &= \sum_{k=0}^{\infty} \frac{(-1)^k}{k+1} x^{k+1} = x - \frac{x^2}{2} + \frac{x^3}{3} - \cdots. \end{aligned}$$

We can notice that the integrated series converges at $x = 1$ and so the convergence is uniform on $[0,1]$. It follows that the representation is valid for $x \in (-1,1]$ but for no other points. In this case we obtained a valid Taylor series expansion by integrating a series expansion that we already knew to be valid. ◀

To study the convergence of a Taylor series in general, let us return to fundamentals. Let f be infinitely differentiable in a neighborhood of c. For $n = 0,1,2,\ldots$ let

$$P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(c)}{k!}(x-c)^k.$$

The polynomial P_n is called the *nth Taylor polynomial* of f at c. The difference $R_n(x) = f(x) - P_n(x)$ is called the *nth remainder function*. In order for the Taylor series about c to converge to f on an interval I, it is necessary and sufficient that $R_n \to 0$ pointwise on I.

Example 10.25: We know that the geometric series represents the function $f(x) = (1-x)^{-1}$ on the interval $(-1,1)$. We could also prove this result by relying on the remainder term. For $x \neq 1$ and $n = 0,1,2,\ldots$ we have

$$\frac{1}{1-x} = 1 + x + x^2 + \cdots + x^n + \frac{x^{n+1}}{1-x}.$$

Here

$$P_n(x) = 1 + x + x + x^2 + \cdots + x^n$$

and

$$R_n(x) = \frac{x^{n+1}}{1-x}.$$

For $|x| < 1$, $R_n(x) \to 0$ as $n \to \infty$. But we have

$$f(x) = P_n(x) + R_n(x)$$

and so the Taylor series for $f(x) = 1/(1-x)$ represents f on the interval $(-1,1)$. For $|x| \geq 1$, the remainder term does not tend to zero. As before, we see that the representation is confined to the interval $(-1,1)$. ◀

In a more general situation than this example we would not have an explicit formula for the remainder term. How should we be able to show that the remainder term tends to zero? For functions that are infinitely differentiable in a neighborhood I of c, the various expressions we obtained in Section 7.12 for the remainder functions R_n can be used. In particular, Lagrange's form of the remainder allows us to write for $n = 0, 1, 2, 3, \ldots$

$$f(x) = P_n(x) + \frac{f^{(n+1)}(z)}{(n+1)!}(x-c)^{n+1},$$

where z is between x and c. With some information on the size of the derivatives $f^{(n+1)}(z)$ we might be able to show that this remainder term tends to zero. The integral form of the remainder term, gives us

$$f(x) = P_n(x) + \frac{1}{n!}\int_c^x (x-t)^n f^{(n+1)}(t)\,dt.$$

Again information on the size of the derivatives $f^{(n+1)}(t)$ might show that this remainder term tends to zero.

Example 10.26: Let us justify the familiar Taylor series for $\sin x$:

$$\sin x = \sum_0^\infty \frac{(-1)^k}{(2k+1)!}x^{2k+1}. \tag{6}$$

The remainder term is not expressible in any simple way but can be estimated by using the Lagrange's form of the remainder. The coefficients

$$\frac{(-1)^k}{(2k+1)!}$$

are easily verified by calculating successive derivatives of $f(x) = \sin x$ and using the formulas

$$a_k = \frac{f^{(k)}(0)}{k!}.$$

To check convergence of the series, apply Lagrange's form for $R_n(x)$: For each $x \in \mathbb{R}$, there exists z such that

$$R_n(x) = \frac{f^{(n+1)}(z)}{(n+1)!}x^{n+1}.$$

Now $|f^{(n+1)}(z)|$ equals either $|\cos z|$ or $|\sin z|$ (depending on n) so, in either case, $|f^{(n+1)}(z)| \leq 1$, and

$$|R_n(x)| \leq |x|^{n+1}/(n+1)!.$$

Since $|x|^{n+1}/(n+1)! \to 0$ as $n \to \infty$ for all $x \in \mathbb{R}$, we can see that the remainder term $|R_n(x)| \to 0$ as $n \to \infty$ for all $x \in \mathbb{R}$. Thus the series representation is completely justified for all real x.

Observe that our estimate for $|R_n(x)|$,

$$|R_n(x)| \leq |x|^{n+1}/(n+1)!,$$

gives also a sense of the *rate* of convergence of the series for fixed x. For example, for $|x| \leq 1$ we find

$$|R_n(x)| \leq 1/(n+1)!.$$

Thus, if we want to calculate $\sin x$ on $(-1,1)$ to within .01, we need take only the first five terms of the series $(n = 4)$ to achieve that degree of accuracy.

Had we used the integral form for $R_n(x)$ we would have obtained a similar estimate. We leave that calculation as Exercise 10.5.1. ◀

10.5.2 Analytic Functions

The class of functions that can be represented as power series is not large. As we have remarked, the class of infinitely differentiable functions is much larger. The terminology that is commonly used for this very special class of functions is given by the following definition.

Definition 10.27: A function f whose Taylor series converges to f in a neighborhood of c is said to be *analytic at c*.

The functions commonly encountered in elementary calculus are generally analytic except at certain obviously nonanalytic points. For example, $|x|$ is not analytic at $x = 0$, and $1/(1-x)$ is not analytic at $x = 1$. These functions fail to have even a first derivative at the point in question. Similarly a function such as $f(x) = |x|^3$ cannot be analytic at $x = 0$ because, while $f'(0)$ and $f''(0)$ exist, $f^{(3)}(0)$ does not. It is not possible to write the complete Taylor series for such a function since some of the derivatives fail to exist.

Even if a function has infinitely many derivatives at a point it need not be analytic there. We would be able to write the complete Taylor expansion but, as we have already noted, the resulting series might not converge to f on any interval. In this connection, it is instructive to work Exercise 10.5.4.

In Example 10.26 we justified the Taylor expansion for $\sin x$. Part of the justification involved the fact that $\sin x$ and all of its derivatives are bounded on the real line. This suggests a general result.

Theorem 10.28: *Let f be infinitely differentiable in a neighborhood I of c. Suppose $x \in I$ and there exists $M > 0$ such that*

$$|f^{(m)}(t)| \leq M$$

for all $m \in \mathbb{N}$ and $t \in [c,x]$ (or $[x,c]$ if $x < c$). Then

$$\lim_{n\to\infty} R_n(x) = 0.$$

Thus, f is analytic at c.

Proof. We prove the theorem for $x > c$. We leave the case $x < c$ as Exercise 10.5.5.

We use the integral form of the remainder (Theorem 7.45), obtaining

$$|R_n(x)| = \left| \frac{1}{n!} \int_c^x (x-t)^n f^{(n+1)}(t)\, dt \right|. \tag{7}$$

Using our hypothesis that $|f^{(m)}(t)| \leq M$ for all $t \in [c,x]$, we infer from (7) that

$$\begin{aligned} |R_n(x)| &\leq \left| \frac{M}{n!} \int_c^x (x-t)^n \, dt \right| \\ &= \left| \frac{M}{n!} \frac{(x-t)^{n+1}}{n+1} \Big|_c^x \right| \\ &= \left| \frac{M}{(n+1)!} (x-c)^{n+1} \right|. \end{aligned}$$

For fixed x and c, $(x-c)$ is just a constant, so

$$\frac{M(x-c)^{n+1}}{(n+1)!} \to 0.$$

Thus $|R_n(x)| \to 0$ and f is analytic at c. ■

Example 10.29: Let us show that the function $f(x) = e^x$ is analytic at $x = 0$. It is infinitely differentiable, but we need to prove more. The fact that f is analytic at $x = 0$ follows from the previous theorem: We choose, say, the interval $[-1,1]$ and note that $|f^{(n)}(x)| = |e^x| \leq e$ for all $x \in (-1,1)$ and $n \in \mathbb{N}$. A similar observation applies to the analyticity of f at any point $c \in \mathbb{R}$. ◀

Exercise 10.5.6 provides another theorem similar to Theorem 10.28.

Exercises

10.5.1 Justify formula (6) for $\sin x$ using the integral form of the remainder $R_n(x)$.

10.5.2 Show that

$$f(x) = \sum_{k=0}^{n} \frac{f^k(0)}{k!} x^k + \frac{x^{n+1}}{n!} \int_0^1 f^{(n+1)}(sx)(1-s)^n \, ds$$

under appropriate assumptions on f.

10.5.3 Show that

$$\int_0^1 f^{(n+1)}(sb)(1-s)^n \, ds \leq \frac{n! f(b)}{b^{n+1}}$$

if f and all of its derivatives exist and are nonnegative on the interval $[0,b]$.

10.5.4 Let

$$f(x) = \begin{cases} 0, & \text{if } x = 0 \\ e^{-1/x^2}, & \text{if } x \neq 0. \end{cases}$$

Prove that f is infinitely differentiable on the real line. Show that $f^{(k)}(0) = 0$ for all $k \in \mathbb{N}$. Explain why the Taylor series for f about $x = 0$ does not represent f in any neighborhood of zero. Is f analytic at $x = c$ for $c \neq 0$?

See Note 253

10.5.5 Prove Theorem 10.28 for $x < c$.

10.5.6 Prove Bernstein's Theorem: If f is infinitely differentiable on an interval I, and $f^{(n)}(x) \geq 0$ for all $n \in \mathbb{N}$ and $x \in I$, then f is analytic on I. Apply this result to $f(x) = e^x$.

10.5.7 Use the results of this section to verify that each of the following functions is analytic at $x = 0$, and write the Taylor series about $x = 0$.

(a) $\cos x^2$

(b) e^{-x^2}

10.5.8 Show that if f and g are analytic functions at each point of an interval (a,b), then so too is any linear combination $\alpha f + \beta g$.

10.6 Products of Power Series

Suppose that we have two power series representations

$$f(x) = \sum_{k=0}^{\infty} a_k (x - x_0)^k$$

and

$$g(x) = \sum_{k=0}^{\infty} b_k (x - x_0)^k$$

valid in the intervals $(-R_f, R_f)$ and $(-R_g, R_g)$, respectively. How should we obtain a power series representation for the product $f(x)g(x)$? We might merely compute all the derivatives of this function and so construct its Taylor series. But is this the easiest or most convenient method? How do we know that such a representation would be valid?

The most direct approach to this problem is to apply here our study of products of series from Section 3.8. We know when such a product would be valid. Indeed, from that theory, we know immediately that

$$f(x)g(x) = \sum_{k=0}^{\infty} c_k (x - x_0)^k$$

would hold in the interval $(-R, R)$, where $R = \min\{R_f, R_g\}$ and the coefficients are given by the formulas

$$c_k = \sum_{j=0}^{k} a_j b_{k-j}.$$

Example 10.30: The product of the series

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

and the series

$$f(x) = \sum_{k=0}^{\infty} a_k x^k$$

gives the representation

$$\frac{f(x)}{1-x} = \sum_{k=0}^{\infty} (a_0 + a_1 + a_2 + \cdots + a_k) x^k.$$

Where would this be valid? ◀

Example 10.31: A representation for the function $e^x \sin x$ might be most easily obtained by forming the product

$$\left(1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots\right)\left(x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \dots\right) = x + x^2 + \frac{1}{6}x^3 + \dots$$

and the series continued as far as is needed for the application at hand. ◀

10.6.1 Quotients of Power Series

Suppose that we have power series representations of two functions

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \text{ and } g(x) = \sum_{k=0}^{\infty} b_k x^k$$

both valid in some interval $(-r, r)$ at least. Can we find a representation of the quotient function $f(x)/g(x)$? Certainly we must demand that $g(0) \neq 0$, which amounts to asking for the leading coefficient in the series for g (the term b_0) not to be zero.

If there is a representation, say a series $\sum_{k=0}^{\infty} c_k x^k$, then, evidently, we require that

$$\frac{\sum_{k=0}^{\infty} a_k x^k}{\sum_{k=0}^{\infty} b_k x^k} = \sum_{k=0}^{\infty} c_k x^k.$$

This merely means that we want

$$\left(\sum_{k=0}^{\infty} b_k x^k\right)\left(\sum_{k=0}^{\infty} c_k x^k\right) = \sum_{k=0}^{\infty} a_k x^k.$$

The conditions for this are known to us since we have already studied how to form the product of two power series. For this to hold the coefficients $\{c_k\}$ (which, at the moment, we do not know how to determine) should satisfy the equations

$$b_0 c_0 = a_0$$

$$b_0 c_1 + b_1 c_0 = a_1$$

$$b_0 c_2 + b_1 c_1 + b_2 c_0 = a_2$$

and, in general,

$$b_0 c_k + b_1 c_{k-1} + b_2 c_{k-2} + \cdots + b_k c_0 = a_k.$$

Since we know all the a_k's and b_k's, we can readily solve these equations, one at a time starting from the first to obtain the coefficients for the quotient series. This algorithm (for that is what it is) for determining the c_k's is precisely "long division." Simply divide formally the expression (the denominator)

$$b_0 + b_1 x + b_2 x^2 + b_3 x^3 + \ldots$$

into the expression (the numerator)

$$a_0 + a_1 x + a_2 x^2 + a_3 x^3 + \ldots$$

and you will find yourself solving exactly these equations in our algorithm.

But what have we determined? We have shown that if there is a series representation for $f(x)/g(x)$, then this method will determine it. We do not, however, have any assurances in advance that there is such a series. We offer the next theorem, without proof, for those assurances. Alternatively, in any computation we could construct the quotient series (all terms!) and determine that it has a positive radius of convergence. That, too, would justify the method although it is not likely the most practical approach.

Theorem 10.32: *Suppose that there are power series representations for two functions*

$$f(x) = \sum_{k=0}^{\infty} a_k x^k \text{ and } g(x) = \sum_{k=0}^{\infty} b_k x^k$$

both valid in some interval $(-r,r)$ at least and that $b_0 \neq 0$. Then there is some positive δ so that the function $f(x)/g(x)$ is analytic in $(-\delta,\delta)$ and a quotient series can be found there.

The proper setting for a proof of Theorem 10.32 is complex analysis, where it is proved that a quotient of complex analytic functions is analytic if the denominator is not zero.

Exercises

10.6.1 Show that if f and g are analytic functions at each point of an interval (a,b), then so too is the product fg.

10.6.2 Under what conditions on the functions f and g on an interval (a,b) can you conclude that the quotient f/g is analytic?
SEE NOTE 254

10.6.3 Using long division, find several terms of the power series expansion of

$$\frac{x+2}{x^2+x+1}$$

centered at $x=0$. What other method would work?

10.6.4 Using long division and the power series expansions for $\sin x$ and $\cos x$, find the first few terms of the power series expansion of $\tan x$ centered at $x=0$. What other method would have given you these same numbers?

10.6.5 Find a power series expansion centered at $x=0$ for the function

$$\frac{\sin 2x}{\sin x}.$$

Did the fact that $\sin x = 0$ at $x=0$ make you modify the method here?

10.6.6 Show that if

$$\frac{1}{\sum_{k=0}^{\infty} b_k x^k} = \sum_{k=0}^{\infty} c_k x^k$$

is valid, then

$$c_k = \frac{(-1)^k}{b_0^{k+1}} \begin{vmatrix} b_1 & b_0 & 0 & 0 & \dots & 0 \\ b_2 & b_1 & b_0 & 0 & \dots & 0 \\ b_3 & b_2 & b_1 & b_0 & \dots & 0 \\ \dots & & & & \dots & \dots \\ b_k & b_{k-1} & b_{k-2} & b_{k-3} & \dots & b_1 \end{vmatrix}.$$

10.7 Composition of Power Series

Suppose that we wished to obtain a power series expansion for the function $e^{\sin x}$ using the two series expansions

$$e^x = 1 + x + \frac{1}{2}x^2 + \frac{1}{6}x^3 + \dots$$

and

$$\sin x = x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \dots.$$

Without pausing to decide if this makes any sense let us simply insert the series for $\sin x$ in the appropriate positions in the series for e^x. Then we might hope to justify that

$$e^{\sin x} = 1 + \left(x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \dots \right)$$
$$+ \frac{1}{2}\left(x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \dots \right)^2 + \frac{1}{6}\left(x - \frac{1}{3}x^3 + \frac{1}{5}x^5 - \dots \right)^3 + \dots$$

and expand, grouping terms in the obvious way, getting (at least for a start)

$$e^{\sin x} = 1 + x + \frac{1}{2}x^2 - \frac{1}{6}x^3 + \dots .$$

Is this method valid?

To justify this method we state (without proof) a theorem giving some conditions when this could be verified. Note that the conditions are as we should expect for a composition of functions $f(g(x))$. The series for $g(x)$ is expanded about a point x_0. That is inserted into a series expanded about the value $g(x_0)$, thus obtaining a series for $f(g(x))$ expanded about the point x_0. The proof is not difficult if approached within a course in complex variables but would be mysterious if attempted as a real variable theorem.

Theorem 10.33: *Suppose that there are power series representations for two functions*

$$g(x) = C + \sum_{k=1}^{\infty} a_k (x - x_0)^k \quad \text{and} \quad f(x) = \sum_{k=0}^{\infty} b_k (x - C)^k$$

both valid in some nondegenerate intervals about their centers. Then there is a power series expansion for

$$f(g(x)) = \sum_{k=0}^{\infty} c_k (x - x_0)^k$$

with a positive radius of convergence whose coefficients can be obtained by inserting the series for $g(x) - C$ into the series for f, that is, by expanding

$$f(g(x)) = \sum_{k=0}^{\infty} b_k \left(\sum_{j=1}^{\infty} a_j (x - x_0)^j \right)^k$$

formally.

Exercises

10.7.1 Under what conditions on the functions f and g on an interval (a,b) can you conclude that the composition $f \circ g$ is analytic?

See Note 255

10.7.2 Find several terms in the power series expansion of $e^{\sin x}$ by a method different from that in this section.

10.7.3 Find several terms in the power series expansion of $e^{\tan x}$ using the method discussed in this section.

10.8 Trigonometric Series

✂ *Enrichment section. May be omitted.*

In this section we present a short introduction to another way of representing functions, namely as trigonometric series or Fourier series. There are deep connections between power series and Fourier series so this theory does belong in this chapter (see Exercise 10.8.1).

The origins of the subject go back to the middle of the eighteenth century. Certain problems in mathematical physics seemed to require that an arbitrary function f with a fixed period (taken here as 2π) be represented in the form of a trigonometric series

$$f(t) = \tfrac{1}{2}a_0 + \sum_{j=1}^{\infty} (a_j \cos jt + b_j \sin jt), \tag{8}$$

and mathematicians such as Daniel Bernoulli, d'Alembert, Lagrange, and Euler had debated whether such a thing should be possible. Bernoulli maintained that this would always be possible, while Euler and d'Alembert argued against it.

Joseph Fourier (1768–1830) saw the utility of these representations and, although he did nothing to verify his position other than to perform some specific calculations, claimed that the representation in (8) would be available for every function f and gave the formulas

$$a_j = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos jt\,dt \quad \text{and} \quad b_j = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin jt\,dt$$

for the coefficients.

While his mathematical reasons were not very strong and much criticized at the time, his instincts were correct, and series of this form with coefficients computed in this way are now known as *Fourier series.* The a_j and b_j are called the *Fourier coefficients* of f.

10.8.1 Uniform Convergence of Trigonometric Series

✂ *Enrichment section. May be omitted.*

For a first taste of this theory we prove an interesting theorem that justifies some of Fourier's original intuitions. We show that if a trigonometric series converges *uniformly* to a function f, then necessarily those coefficients given by Fourier are the correct ones.

Theorem 10.34: *Suppose that*

$$f(t) = \tfrac{1}{2}a_0 + \sum_{j=1}^{\infty} (a_j \cos jt + b_j \sin jt), \tag{9}$$

with uniform convergence on the interval $[-\pi,\pi]$. *Then it follows that the function* f *is continuous and the coefficients are given by Fourier's formulas:*

$$a_j = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\cos jt\,dt \quad \textit{and} \quad b_j = \frac{1}{\pi}\int_{-\pi}^{\pi} f(t)\sin jt\,dt$$

Proof. Fix $j \geq 1$, choose $n > j$ and write

$$s_n(t) = \tfrac{1}{2}a_0 + \sum_{k=1}^{n} (a_k \cos kt + b_k \sin kt)$$

that is, the partial sums of the series. A straightforward, if tiresome, calculation shows that for $j \geq 1$, and for $n > j$,

$$\int_{-\pi}^{\pi} s_n(t) \cos jt \, dt = \int_{-\pi}^{\pi} a_j (\cos jt)^2 \, dt = a_j \pi. \tag{10}$$

This is, remember, just a finite sum. The orthogonality relations in Exercise 10.8.3 assist in this computation.

We are assuming that $s_n \to f$ uniformly and so it follows too, since $\cos jt$ is bounded that $s_n(t) \cos jt \to f(t) \cos jt$ uniformly for $t \in [-\pi, \pi]$. It follows, since all functions here are continuous, that

$$\lim_{n\to\infty} \int_{-\pi}^{\pi} s_n(t) \cos jt \, dt = \int_{-\pi}^{\pi} f(t) \cos jt \, dt.$$

In view of (10) this proves the formula for a_j and $j \geq 1$. The formulas for a_0 and b_j for $j \geq 1$ can be obtained by an identical method. ■

10.8.2 Fourier Series

✂ *Enrichment section. May be omitted.*

Emboldened by the theorem we have just proved we make a dramatic move, the same move that Fourier made. We start with the function f (not the series) and construct a trigonometric series by using these coefficient formulas.

Note the twist in the logic. *If* there is a trigonometric series converging uniformly to a continuous function f, then it would have to be given by the formulas of Theorem 10.34. Why not start with the series even if we have no knowledge that the series will converge uniformly, even if we do not know whether it will converge uniformly to the function we started with, indeed even if the series diverges?

Definition 10.35: Let f be a continuous function[3] on the interval $[-\pi, \pi]$ and let

$$a_j = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \cos jt \, dt \quad \text{and} \quad b_j = \frac{1}{\pi} \int_{-\pi}^{\pi} f(t) \sin jt \, dt.$$

Then the series

$$\tfrac{1}{2} a_0 + \sum_{j=1}^{\infty} (a_j \cos jt + b_j \sin jt) \tag{11}$$

is called the *Fourier series* of f.

There is a mild understanding here that the series should be somehow related to f and there is a hope that the series can be used as a "representation" of f. But, in general, uniform convergence is out of the question. Indeed, even pointwise convergence is too much to hope for. To emphasize that this relation is not one of equality, we usually write

$$f(t) \sim \tfrac{1}{2} a_0 + \sum_{j=1}^{\infty} (a_j \cos jt + b_j \sin jt).$$

[3] We could assume merely that f is Riemann integrable, but in fact our theorems will concern only continuous functions. The study of Fourier series is considerably limited using only Riemann integration (instead of the more general Lebesgue integral).

Exercises

10.8.1 Let $f(z) = \sum_{k=0}^{\infty} \alpha_k z^k$ be a complex power series with a radius of convergence larger than 1. By setting $z = e^{it}$ find a connection between complex power series and trigonometric series.

10.8.2 Explain why it is that for any integrable function f we can claim that the integrals defining the Fourier coefficients of f exist.

10.8.3 Check the so-called *orthogonality relations* by computing that for integers $k \neq j$ and all i

$$\int_{-\pi}^{\pi} \sin(kt)\sin(jt)\,dt = 0, \quad \int_{-\pi}^{\pi} \cos(kt)\sin(it)\,dt = 0,$$

and

$$\int_{-\pi}^{\pi} \cos(kt)\cos(jt)\,dt = 0.$$

10.8.4 Check that for integers i, $k \neq 0$,

$$\int_{-\pi}^{\pi} (\sin kt)^2\,dt = \pi \quad \text{and} \quad \int_{-\pi}^{\pi} (\cos it)^2\,dt = \pi.$$

10.8.3 Convergence of Fourier Series

✂ *Enrichment section. May be omitted.*

The theory of Fourier Series would have a much simpler, if less fascinating, development if the Fourier series of every continuous function converged uniformly to the original function. Not only is this false, but the Fourier series of a continuous function can diverge at a large set of points. This leaves us with a serious difficulty. The Fourier series of a function is expected to represent the function, but how? If it does not converge to the function, how can it be used as a representation?

There is a mistake in our reasoning. We know that if a series converges to a function in suitable ways, then the function may be integrated and differentiated by termwise integration and differentiation of the series. But it is possible that a series may be manipulated in these ways *even if the series diverges* at some points. A representation need not be a pointwise or uniform representation to be useful.

In our next theorem we show that the Cesàro sums of the Fourier series of a suitable function do converge uniformly to the function even if the series itself is divergent. You should review the topic of Cesàro summability in Section 3.9.1. A young Hungarian mathematician, Leopold Fejér (1880–1959), obtained this theorem in 1900.

Theorem 10.36 (Fejér) *Let f be a continuous function on $[-\pi,\pi]$ for which $f(-\pi) = f(\pi)$. Then the sequence of Cesàro means of the partial sums of the Fourier series for f converges uniformly to f on $[-\pi,\pi]$.*

Proof. Throughout the proof we may consider that f is defined on all of $\mathbb{R}$ and is 2π-periodic. We write

$$s_n(x) = \tfrac{1}{2}a_0 + \sum_{k=1}^{n} (a_k \cos kx + b_k \sin kx)$$

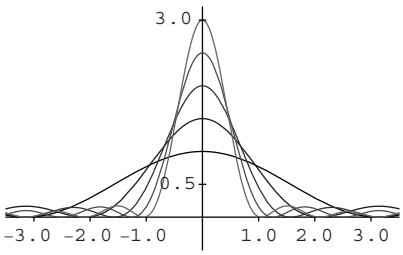


Figure 10.1. Fejér kernel $K_n(t)$ for $n = 1, 2, 3, 4$, and 5 on $[-\pi, \pi]$.

for the partial sums of the Fourier series of f (this means the coefficients a_j, b_j are determined by using Fourier's formulas). Then we write

$$\sigma_n(x) = \frac{s_0(x) + s_1(x) + s_2(x) + \cdots + s_n(x)}{n+1}$$

for the sequence of averages (Cesàro means).

Our task is to prove that $\sigma_n \to f$ uniformly. Looking back we see that each $\sigma_n(x)$ is a finite sum of terms $s_k(x)$ and each $s_k(x)$ is a finite sum of terms involving a_j, b_j. In turn, each of these terms is expressible as an integral involving f and sin's and cos's. Thus after some considerable but routine computations, we arrive at a formula

$$\sigma_n(x) = \frac{1}{\pi} \int_{-\pi}^{\pi} \frac{1}{2} \left(f(x+t) + f(x-t) \right) K_n(t)\, dt$$

or the equivalent formula

$$\sigma_n(x) = \frac{1}{\pi} \int_{0}^{\pi} \left(f(x+t) + f(x-t) \right) K_n(t)\, dt. \tag{12}$$

Here K_n is called the *Fejér kernel* and for each n,

$$K_n(t) = \frac{1}{2(n+1)} \left(\frac{\sin\left(\frac{1}{2}(n+1)t\right)}{\sin \frac{1}{2}t} \right)^2.$$

You can just accept the computations for the purposes of our short introduction to the subject.

The Fejér kernel of order n enjoys the following properties, each of which is evident from its definition:

1. Each $K_n(t)$ is a nonnegative, continuous function.

2. For each n,
$$\frac{1}{\pi} \int_{-\pi}^{\pi} K_n(t)\, dt = \frac{2}{\pi} \int_{0}^{\pi} K_n(t)\, dt = 1.$$

3. For each n and $0 < |t| < \pi$,
$$0 \le K_n(t) \le \frac{\pi}{(n+1)t^2}.$$

Figure 10.1 illustrates the graph of this function for $n = 1, 2, 3, 4$, and 5.

Let $\varepsilon > 0$, and choose $\delta > 0$ so that

$$|f(x+t) + f(x-t) - 2f(x)| < \varepsilon$$

for every $0 \le t \le \delta$. This uses the uniform continuity of f. We note that

$$\frac{2}{\pi}\int_0^\pi f(x)K_n(t)\,dt = f(x)$$

by using property 2. Thus we have

$$\begin{aligned}|\sigma_n(x) - f(x)| &\le \frac{1}{\pi}\int_0^\pi |f(x+t)+f(x-t)-2f(x)|\,K_n(t)\,dt \\ &\le I_1 + I_2,\end{aligned}$$

where I_1 is the integral taken over $[0,\delta]$ and I_2 is the integral taken over $[\delta,\pi]$. Since K_n is nonnegative, we did not need to keep it inside the absolute value in the integral. The part I_1 will be small (for all n) because the expression in the absolute values is small for t in the interval $[0,\delta]$. The part I_2 will be small (for large n) because of the bound on the size of K_n for t away from zero in property 3. Here are the details: For I_1 we have

$$I_1 \le \frac{\varepsilon}{\pi}\int_0^\delta K_n(t)\,dt \le \varepsilon.$$

For I_2, let

$$\kappa_n = \sup\{K_n(t) : \delta \le t \le \pi\},$$

and note that property 3 supplies us with the fact that $\kappa_n \to 0$ as $n \to \infty$. Now we have

$$I_2 \le \frac{\kappa_n \varepsilon}{\pi}\int_\delta^\pi \left(|f(x+t)| + |f(x-t)| + 2|f(x)|\right)\,dt$$

so that we can make I_2 as small as we please by choosing n large enough.

It follows, since ε and x are arbitrary, that $\lim_{n\to\infty}\sigma_n(x) = f(x)$, uniformly for $x \in [-\pi,\pi]$ as required. ■

Exercises

10.8.5 Let $s_n(x)$ be the sequence of partial sums of the Fourier series for a 2π-periodic integrable function f. Show that

$$s_n(x) = \frac{1}{\pi}\int_{-\pi}^\pi \frac{1}{2}\left(f(x+t)+f(x-t)\right)D_n(t)\,dt$$

and

$$s_n(x) = \frac{1}{\pi}\int_0^\pi \left(f(x+t)+f(x-t)\right)D_n(t)\,dt,$$

where

$$D_n(t) = \frac{1}{2} + \sum_{k=1}^n \cos kt$$

is called the Dirichlet kernel.

Figure 10.2 illustrates the graph of this function for $n = 1$, 3, and 7. It should be contrasted with Figure 10.1.

SEE NOTE 256

10.8.6 Establish the following properties of the Dirichlet kernel $D_n(t)$ for each n:

(a) $D_n(t)$ is a continuous, 2π-periodic function.

(b) $D_n(t)$ is an even function.

(c) $\dfrac{1}{\pi}\displaystyle\int_{-\pi}^\pi D_n(t)\,dt = \frac{2}{\pi}\int_0^\pi D_n(t)\,dt = 1.$

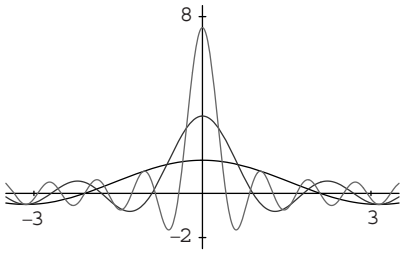


Figure 10.2. Dirichlet kernel $D_n(t)$ for $n = 1$, 3, and 7 on $[-\pi, \pi]$.

(d) $D_n(t) = \dfrac{\sin\left(n+\frac{1}{2}\right)t}{2\sin\frac{1}{2}t}$.

(e) $D_n(0) = n+\frac{1}{2}$.

(f) For all t, $|D_n(t)| \le n+\frac{1}{2}$.

(g) For all $0 < |t| < \pi$, $|D_n(t)| \le \frac{\pi}{2|t|}$.

10.8.7 Let

$$K_n(t) = \frac{1}{n+1}\sum_{j=0}^{n} D_j(t),$$

where D_j are the Dirichlet kernels. Show that the formula for the averages σ_n given in the proof of Theorem 10.36 is correct.

10.8.4 Weierstrass Approximation Theorem

✂ *Enrichment section. May be omitted.*

Fejér's theorem allows us to prove the famous Weierstrass approximation theorem. Note that a consequence of Fejér's theorem is that continuous, 2π-periodic functions can be uniformly approximated by trigonometric polynomials. The Weierstrass theorem asserts that continuous functions on a compact interval can be uniformly approximated by ordinary polynomials.

Theorem 10.37 (Weierstrass) *Let f be a continuous function on an interval $[a,b]$, and let $\varepsilon > 0$. Then there is a polynomial*

$$g(x) = \alpha_n x^n + \alpha_{n-1}x^{n-1} + \cdots + \alpha_1 x + \alpha_0$$

so that

$$|f(x) - g(x)| < \varepsilon$$

for all $x \in [a,b]$.

Proof. It is more convenient for this proof to assume that $[a,b] = [0,1]$. The general case can be obtained from this.

Let f be a continuous function on $[0,1]$, let $\varepsilon > 0$, and write

$$F(t) = f(|\cos t|).$$

Then F is a continuous, 2π-periodic function and can be approximated by a trigonometric polynomial within ε. This is because, in view of Theorem 10.36, for large enough n the Cesàro means $\sigma_n(F)$ are uniformly close to F.

Since F is even [i.e., $F(t) = F(-t)$] we can figure out what form that trigonometric polynomial may take. All the coefficients b_k involving $\sin kt$ in the Fourier series for F must be zero. Thus when we form the averages of the partial sums we obtain only sums of cosines. Consequently, we can find $c_0, c_1, c_2, \ldots c_n$ so that

$$\left| F(t) - \sum_0^n c_j \cos jt \right| < \varepsilon \tag{13}$$

for all t. Each $\cos jt$ can be written using elementary trigonometric identities as $T_j(\cos t)$ for some jth-order (ordinary) polynomial T_j, and so, by setting $x = \cos t$ for any $x \in [0,1]$, we have

$$\left| f(x) - \sum_0^n c_j T_j(x) \right| < \varepsilon,$$

which is exactly the polynomial approximation that we need. ■

The polynomials T_j that appear in the proof are well known as the Chebychev polynomials and are easily generated (see Exercise 10.8.9). They are named after the Russian mathematician Pafnuty Lvovich Chebychev (1821–1894).

Exercises

10.8.8 Show that once Theorem 10.37 is proved for the interval $[0,1]$ it can be deduced for any interval $[a,b]$.

10.8.9 Define the Chebychev polynomials by requiring T_j to be a polynomial so that

$$\cos jt = T_j(\cos t)$$

identically. Show that $T_0(x) = 1$, $T_1(x) = x$, and

$$T_n(x) = 2xT_{n-1}(x) - T_{n-2}(x).$$

Generate the first few of these polynomials.

10.8.10 Show that Theorem 10.37 can be interpreted as asserting that for any continuous function on an interval $[a,b]$ there is a sequence of polynomials p_n converging to f uniformly on $[a,b]$.

10.8.11 Does Exercise 10.8.10 also imply that there must be a power series expansion converging to f uniformly on $[a,b]$?

10.8.12 Let f be a continuous function on an interval $[a,b]$, and let $\varepsilon > 0$. Show that there must exist a polynomial p with rational coefficients so that

$$|f(x) - p(x)| < \varepsilon$$

for all $x \in [a,b]$.

See Note 257

10.8.13 Let $f : \mathbb{R} \to \mathbb{R}$ be a continuous function and let $\varepsilon > 0$. Must there exist a polynomial p so that $|f(x) - p(x)| < \varepsilon$ for all $x \in \mathbb{R}$?

See Note 258

10.8.14 Let $f : (0,1) \to \mathbb{R}$ be a continuous function and let $\varepsilon > 0$. Must there exist a polynomial p so that $|f(x) - p(x)| < \varepsilon$ for all $x \in (0,1)$?

See Note 259

10.8.15 Let $f : [0,1] \to \mathbb{R}$ be a continuous function with the property that

$$\int_0^1 f(x)x^n \, dx = 0$$

for all $n = 0, 1, 2, \ldots$. What can you conclude about the function f?

See Note 260